ASSC-RBSF: Enhanced RBSF via Joint Adaptive Subband Spectral Cancellation for RFI Artifact Detection and Mitigation in SLC SAR Images

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***Abstract*—In complex electromagnetic environments, the synthetic aperture radar (SAR) is severely affected by radio frequency interference (RFI) from other communication systems, thus prompting the development of multiple RFI mitigation methods. The robust block subspace filtering (RBSF), an improved version of block subspace filtering (BSF), exhibits robustness removing RFI artifacts from single-look complex (SLC) SAR images under strong scattering backgrounds. However, the absence of artifact detection and a suboptimal strategy for RFI estimation leads to performance degradation with complex RFI types across diverse scenarios. This article proposes an advanced variant of RBSF via joint adaptive subband spectral cancellation (ASSC-RBSF) to address the above limitations, thereby improving the applicability of the algorithm in practical applications. Specifically, the proposed method first obtains an approximate estimation of the ground-truth amplitude through the ASSC framework, and then, a change detection operator is utilized to detect and localize RFI artifacts in both two dimensions. Subsequently, dynamic segmentation thresholds are derived based on the amplitude estimation results to determine the RFI subspace dimension for corresponding subblocks, ultimately enabling the reconstruction and separation of RFI artifacts in batches. Comprehensive experiments demonstrate that the use of ASSC enhances the robustness of the RBSF algorithm.**

***Index Terms*—Change detection, radio frequency interference (RFI), robust block subspace filtering (RBSF), subband spectral cancellation (SSC), synthetic aperture radar (SAR).**

I. Introduction

A. Background

SPACEBORNE synthetic aperture radar (SAR), an active microwave remote sensing modality, ensures the persistent all-weather Earth observation, demonstrating the indispensable value in agricultural remote sensing, military reconnaissance, natural disaster monitoring, and other geoscientifically significant scenarios [1], [2], [3], [4]. In recent decades, the spaceborne SAR has undergone the rapid technological advancement, with numerous national space agencies deploying and operating a substantial fleet of SAR satellites. To accommodate the diverse application demands and environmental scenarios, these systems are typically configured to operate within specific frequency bands. In accordance with the radio regulations of the International Telecommunication Union (ITU), the primary frequency bands employed by spaceborne SAR satellites include L, C, S, and X bands, each tailored to distinct remote sensing objectives [6], [7], [8], [9].

The rapid proliferation of global radio technologies has led to increased spectrum sharing among electromagnetic devices [10]. Consequently, SAR satellites are increasingly vulnerable to unintentional radio frequency interference (RFI) from co-channel emitters, such as AM/FM broadcast stations, cellular base stations, and satellite communication systems. Upon reception, RFI distorts the impulse response of SAR signals, degrades the image resolution, and impairs downstream applications, including image interpretation and engineering analysis [11].

A notable increase in RFI-related incidents affecting spaceborne SAR satellites has been witnessed in recent years [12]. As illustrated in Fig. 1, the distribution map of Sentinel-1A satellite's global C-band RFI probability from 2022 to 2025 demonstrates a significant expansion in both the geographic extent and occurrence likelihood of RFI. This trend underscores the critical need for robust methods to mitigate RFI impacts on SAR data quality and ensure reliable high-fidelity imaging in increasingly congested electromagnetic environments.

B. Previous Work

In recent years, numerous RFI mitigation methods have been proposed, broadly categorized into three classes: parametric, semiparametric, and nonparametric approaches. Parametric methods isolate RFI through the mathematical modeling and parameter estimation, typically effective for single-type RFI mitigation. Miller et al. [13] leveraged the prior knowledge of narrowband interference (NBI) to estimate sinusoidal model parameters via least squares. Liu et al. [14] proposed a joint iterative adaptive (IAA) method combined with the orthogonal subspace projection in 2018, significantly enhancing the mitigation accuracy and efficiency of time-varying RFI. In 2020, Yang et al. [15] further advanced

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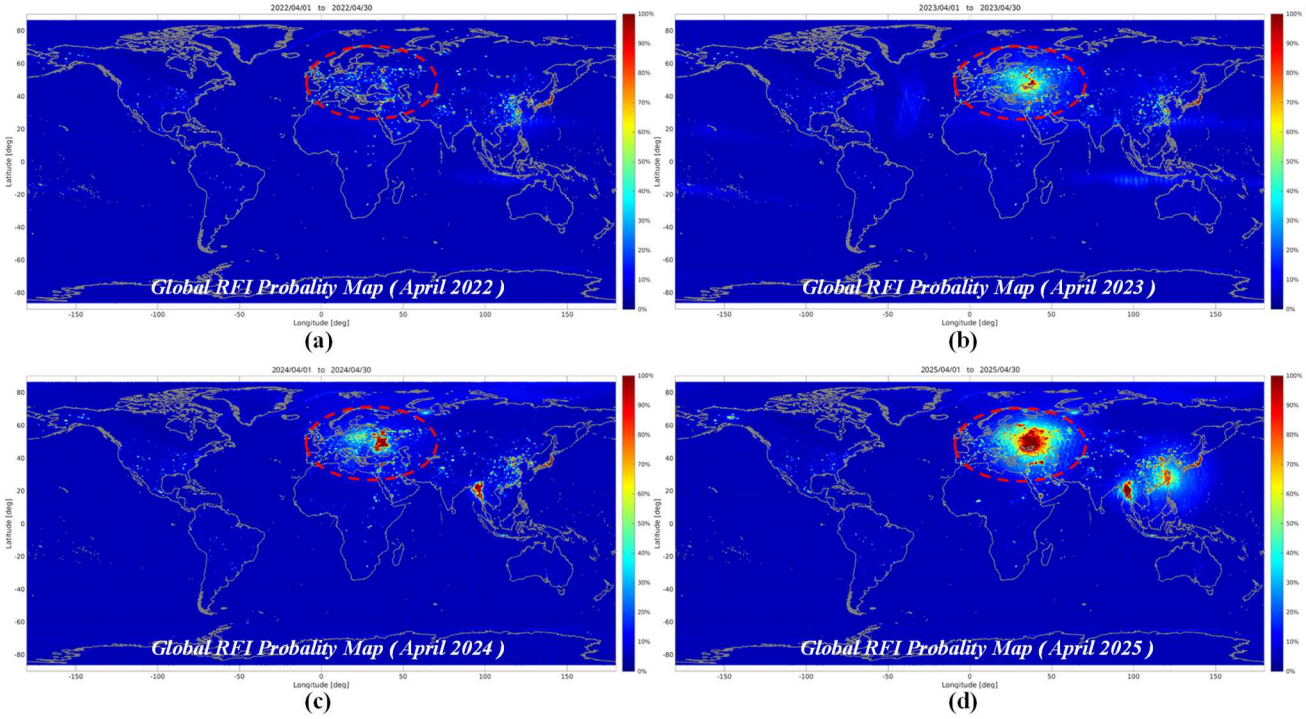


Fig. 1. Global RFI probability distribution map of Sentinel-1A spaceborne SAR system in (a) April 2022, (b) April 2023, (c) April 2024, and (d) April 2025.

the IAA framework through the 2-D spectrum analysis (2D-SPECANs) to address the complex broadband RFI. Han et al. [16] proposed a RFI mitigation method via instantaneous frequency estimation and regularized time–frequency filtering, enabling the precise wideband interference (WBI) component extraction in the time–frequency domain. Mao et al. [17] developed a novel group parameter method to jointly refine RFI and target models, achieving effective RFI mitigation with scene detail preservation via adaptive regularization.

In contrast, nonparametric methods leverage the statistical characteristics of RFI signals and SAR echoes to realize effective RFI extraction and mitigation. As a simple yet mitigation technique, the notch filtering method has seen extensive engineering deployment in recent years. Prominent implementations include time-domain notch (TNF) [18], frequency-domain notch (FNF) [19], time–FNF (TFNF) [20], and their modified variants. However, notch filters inherently cause collateral loss of useful signals during mitigation, a limitation that amplifies with increasing RFI bandwidth. To mitigate RFI while preserving the integrity of the SAR signal, eigendecomposition-based methods for RFI extraction have gained significant traction. Zhou et al. [21] first introduced this paradigm and proposed an eigenvalue subspace projection (ESP) method for NBI mitigation. By decomposing the echo matrix into eigenspaces, this method effectively isolates RFI components from the signal. Subsequently, their team developed more eigendecomposition-based frameworks, including independent principal component analysis (ICA) [22], complex empirical mode decomposition (CEMD) [23], and independent subspace analysis (ISA) [24]. Li et al. [25] and Hu [26] advanced the ESP framework introducing a 1-D azimuth dynamic threshold (1DDT-ESP) and a 2-D azimuth-range dynamic threshold (2DDT-ESP), respectively, which achieved notable efficacy in mitigating ultrawideband (UWB) mutual RFI between SAR satellites.

The semiparametric method integrates the strengths of parametric and nonparametric approaches, where the parametric component is characterized by finite-dimensional hyperparameters, and the nonparametric component is modeled by unknown functions [27]. RFI separation and mitigation are achieved through hyperparameter optimization. Nguyen et al. [28] introduced sparse signal modeling for UWB SAR and first proposed the concept of semiparametric methodology. Liu et al. [29] later employed sparse representation to mitigate WBI and reconstruct SAR signals in the time–frequency domain. Since 2018, Huang et al. [30], [31], [32], [33], [34] have systematically developed a series of "low-rank + sparse" models, incorporating diverse optimization schemes to enable efficient RFI separation and advance the theoretical framework. Mao et al. [35] proposed an adaptive notch semiparametric (ANSP) method that fuses robust principal component analysis (RPCA) with the frequency-domain notching to mitigate RFI under low signal-to-interference-plus-noise ratio (SINR) conditions. Li et al. [36] comprehensively analyzed the factors influencing the low-rank characteristics of RFI and established a general RFI model. By integrating low-rank feature recovery (LRFR) technology, this model enables effective suppression of RFI in complex scenarios. What's more, learning-based RFI mitigation methods have also received increasing attention, which focus on detecting complex or irregular RFI artifacts and in preserving weak targets [40], [41], [42].

The above three methods all achieve RFI mitigation in zero-level SAR echoes. However, mainstream SAR systems primarily only provide level-1 and higher SAR products, which

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limits the migration of existing methods to SAR images. In recent years, researchers have started addressing this challenge by developing image-level RFI mitigation methods. Feng et al. [40] proposed a robust and efficient subband spectral cancellation (SSC) method based on the holographic characteristics of SAR imaging, which achieves the RFI detection and removal in single-look complex (SLC) SAR images. Yang et al. [41] improved SSC by introducing an RFI detector based on the moment kurtosis coefficient, which enhanced the accuracy of subband division. Li et al. [42] further extended the SSC method by integrating successive cancellation and data accumulation, addressing the performance degradation during the RFI mitigation under a high-interference-signal bandwidth ratio (ISBR). However, the spectral cancellation of SSC causes loss of phase information in the SLC data. To address this limitation, in 2019, Yang et al. [43] verified the low-rank characteristics of RFI in SAR images and adapted the RPCA method for separating RFI artifacts. Tao et al. [44] combined multitemporal analysis with RPCA, modeling time-invariant ground-truth as low-rank attributes to realize RFI artifact extraction in time-series SAR imagery. Yang et al. [45] constructed a generic subspace model revealing the low-dimensional structure of the RFI subspace and proposed a block subspace filtering (BSF) method for direct removal of RFI artifacts in SLC SAR images. Yang et al. [46] developed a robust BSF (RBSF) through the constant false alarm rate (CFAR) detector to address the performance degradation caused by strong point scatterers, along with an adaptive RFI subspace estimation strategy, which outperforms the efficiency of the original BSF algorithm.

In particular, RBSF prevents block filters from improperly removing strong point scatterers with RFI by using a CFAR detector to identify and restore them after filtering. In addition, an adaptive RBSF method is designed, where the RFI subspace dimension is selected by optimizing image contrast and entropy, effectively improving the adaptivity of the filtering process.

However, the RBSF method needs to predetermine the local area, where RFI artifacts may exist before filtering, but due to the lack of preprocessing steps for RFI artifact detection, the efficiency of this method cannot be reliably guaranteed in practical engineering applications. Moreover, the strategy of adaptive optimizing RFI subspace relies on differences between RFI artifacts and the background in indicator responses. When these prerequisites are difficult to satisfy due to various scenarios or complex RFI types, the performance of the adaptive RBSF method degrades to varying degrees.

C. Solution and Contributions of This Article

To address the above limitations of the RBSF method, this article presents an advanced variant of RBSF via joint adaptive SSC (ASSC-RBSF), which overcomes the performance degradation across diverse scenarios and complex RFI types through a hierarchical framework. First, the range spectrum is partitioned into subband via fast Fourier transform (FFT) to isolate RFI with varying bandwidths, followed by continuous cancellation and data accumulation to adaptively separate RFI and derive approximate ground-truth amplitude estimation. Subsequently, the change detection is conducted through comparing the aforementioned result with the original image, enabling 2-D localization of RFI artifacts. Then, the target filtering region is delineated, and block-specific adaptive segmentation thresholds are extracted from amplitude estimation result to optimize the RFI subspace dimension. Finally, the RFI artifact component is reconstructed and separated, thereby generating an RFI-free image.

The main contributions of this article are summarized as follows.

1) An enhanced version of RBSF [46], namely, ASSC-RBSF, based on a three-step process combining SCDA with a change detection operator, is proposed. Unlike RBSF, the proposed method enables accurate detection and 2-D localization of RFI artifacts while maintaining high performance of RFI mitigation across diverse scenarios and complex RFI types.
2) A novel subspace dimension optimization strategy, integrating with ground-truth amplitude estimation, which can adaptively reconstruct the component of RFI artifact, is proposed. In contrast to the metric-based strategy employed in RBSF, the proposed strategy achieves higher reconstruction precision of RFI artifact and superior computational efficiency.
3) Extensive experiments are conducted using spaceborne SAR SLC data with different frequency bands: C-band data from Radarsat-1 and Sentinel-1A, and L-band data from ALOS-2, covering various RFI types and diverse scenarios. Specifically, the multiple factors causing the performance degradation of RBSF are discussed using simulated data. Thereafter, the effectiveness and superiority of the proposed method are further demonstrated through the measured data.

D. Organization of This Article

The remainder of this article is organized as follows. Section II covers the RFI subspace models and mitigation limitations via RBSF. Section III offers the fundamental theories underlying the proposed method. Section IV provides a detailed explanation of the proposed methodology. Section V presents the experimental results and performance analysis of the proposal. Section VI concludes this article.

II. Problem Formulation

In this section, we introduce the background of the RBSF method, including the RFI subspace model in SAR images and its basic theory, and discuss its limitations and potential factors that cause the performance degradation.

A. RFI Subspace Model in SAR Images

As illustrated in Fig. 2, RFI penetrates the SAR system via direct or scattered wave propagation. Generally, RFI observed in SAR signals can be categorized based on key parameters such as signal-to-interference ratio (SIR), ISBR, and modulation scheme [47]. As mentioned in [45], for complex-type RFI received in SAR echo, it can be expressed as the summation of

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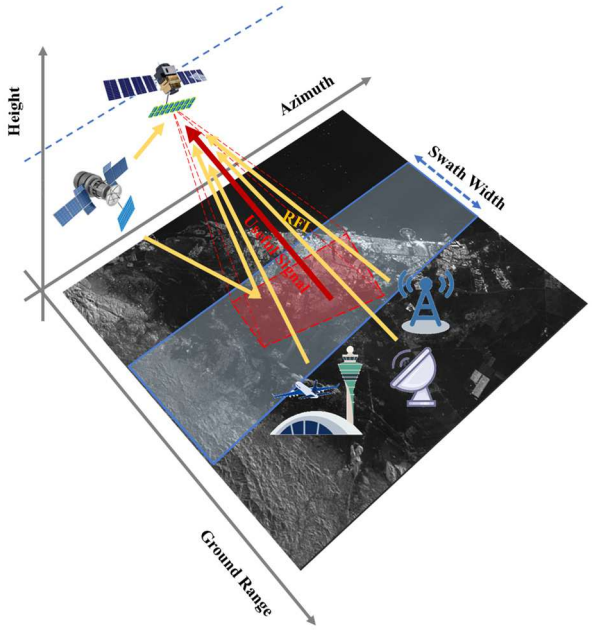


Fig. 2. Typical sources and their associated geometric models.

multiple signal components decoupled in range and azimuth dimensions, as shown in the following equation:

$$R_{\mathrm{RFI}}(\eta,t)=\sum_{i=1}^{L}y_i(\eta,t)=\sum_{i=1}^{L}a_i(\eta)r_i(t) \tag{1}$$

where L represents the number of RFI components. $a_i(\eta)$ and $r_i(t)$ denote the RFI components corresponding to the azimuth time η and range time t, respectively.

Furthermore, the imaging process of the SAR system can be modeled as a 2-D pulse compression implemented by a time-varying matched filter, which is expressed as follows:

$$r(\eta',t')=\sum_{i=1}^{L}\iint a_l(\eta)r_l(t)\,\overline{h}(\eta,t;\eta',t')\,d\eta dt \tag{2}$$

where $\overline{h}$ represents the 2-D matched filter and η' and t' denote the time matching points in the azimuth and range directions, respectively.

B. Basics of the BSF Algorithm

The BSF method proposed by Yang et al. [45] models the RFI artifact by leveraging its low-dimensional characteristics in the image domain, which can be approximately formulated as follows:

$$R_{\mathrm{RFI}}(\eta',t')\approx\sum_{m=1}^{M}\sum_{n=1}^{N}\left[a_{m,n}(\eta')\right]*\left[r_{m,n}(t')\right] \tag{3}$$

where $a_{m,n}(\eta')$ and $r_{m,n}(t')$ represent the RFI components in the azimuth time η' and t', respectively, with M and N being the corresponding feature dimensions.

Based on the above RFI subspace modeling and its low-rank characteristics, the RFI artifact can be expressed as a linear superposition of the ground-truth component and the RFI component, which is defined as follows:

$$\mathbf{S}=\mathbf{X}+\mathbf{I} \tag{4}$$

where $\mathbf{X}$ denotes the RFI-free image, $\mathbf{I}$ represents the RFI artifacts dispersed throughout the background, and $\mathbf{S}$ signifies the RFI-contaminated image. Assuming that the elements of $\mathbf{X}$ are independent identically distributed (i.i.d.) Gaussian, the covariance matrix of $\mathbf{S}$ can be constructed and decomposed through the eigendecomposition as follows:

$$\mathbf{R}=\mathbf{S}\mathbf{S}^{\mathrm{T}}=\sum_{k=1}^{M}\lambda_k e_k e_k^{\mathrm{H}} \tag{5}$$

where k is the index of the eigenvalues corresponding to the RFI response, M represents the total number of eigenvalues, and e_k is the eigenvector corresponding to the eigenvalue λ_k. Finally, the reconstructed RFI component $\mathbf{I}$ is separated to derive a clean image as the RFI-mitigated result.

C. Development of a Robust BSF Algorithm

To address the performance degradation of the BSF method in processing images with strong point scatterers, the RBSF method proposed in [46] incorporates a CFAR-based detector, defined as follows:

$$D[x,y]=\begin{cases}1, & I[x,y]\geq T[x,y]\\ 0, & I[x,y]<T[x,y]\end{cases} \tag{6}$$

where $I[x,y]$ denotes the pixel value at the corresponding position x and y. $D=1$ indicates that the pixel is detected as a strong scattering point; otherwise, it is a background pixel. T represents the dynamic threshold derived from the CFAR operator, which is defined as follows:

$$T[x,y]=L\left(P_{\mathrm{fa}}^{-\frac{1}{L}}-1\right)\left(\bar{\sigma}[x,y]\right) \tag{7}$$

where L represents the number of pixels around the pixel (x,y), P_{fa} denotes the CFAR, and $\bar{\sigma}$ is the mean intensity of the local pixel values. The SAR image is then masked through the detection results. After removing RFI artifacts using the BSF method, the above procedure is repeated to recover the strong point targets within the scene.

In addition, considering that the RFI response degrades the contrast and entropy of the SAR image to a certain extent, a strategy is proposed to adaptively optimize the RFI subspace dimension, which is achieved by solving the following convex optimization problems:

$$\hat{K}_{\mathrm{opt}}=\arg\max_{K}C\left(\hat{\mathbf{I}}=\left|\hat{\mathbf{X}}\right|^2=\left|\mathbf{S}-\tilde{\mathbf{U}}_{1:K}\tilde{\mathbf{S}}\right|^2\right) \tag{8}$$

$$\hat{K}_{\mathrm{opt}}=\arg\min_{K}E\left(\hat{\mathbf{I}}=\left|\hat{\mathbf{X}}\right|^2=\left|\mathbf{S}-\tilde{\mathbf{U}}_{1:K}\tilde{\mathbf{S}}\right|^2\right) \tag{9}$$

where C and E denote the contrast and image entropy, respectively. $\hat{\mathbf{X}}$ represents the mitigated result after removing the RFI subspace $\tilde{\mathbf{U}}_{1:K}\tilde{\mathbf{S}}$, which is constructed through K largest eigenvalues. When the two metrics attain global optimality, the estimation accuracy of RFI reaches a maximum.

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D. Performance Decline of RBSF Under Complex RFI Types and Diverse Scenarios

The performance decline of RBSF in RFI mitigation under complex RFI types and diverse scenarios in practical application can be explained via the following analysis. Given that the metric-based optimization strategy has two prerequisites [46]: 1) the RFI response itself has high entropy or low contrast and 2) the SIR should be appropriately small. This implies that the RFI response characteristics must differ significantly from those of the ground-truth to enable effective removal of RFI artifacts. However, considering the diversity of indicator responses between RFI artifacts and ground-truth background, the above premise may not be satisfied. Fig. 3 illustrates a Sentinel-1A SAR image acquired in Interferometric Wide swath (IW) mode over the San Francisco Bay Area on May 31, 2025, covering diverse land cover types including mountainous terrain, agricultural fields, seaport complexes, and urban centers. Obviously, the indicator response of cropland area is more similar to those of RFI artifacts, which will reduce the estimation accuracy of the RFI subspace dimension. Also, the optimization problem solved in the aforementioned strategy involves extra computational cost, which compromises the algorithm's efficiency in practical applications. Detailed experimental validation and analysis will be provided in Sections V and VI.

III. Related Works

A. NBI Mitigation Based on SSC

Since the imaging process does not alter the bandwidth of the SAR signal, RFI retains its frequency-domain characteristics in the SLC data [6], [48]. To clarify the SSC method, consider two subbands as an example, the power spectrum of the SAR data can be expressed as follows:

$$|S_t|^2 \approx |S_e|^2 + |S_i|^2 \tag{10}$$

where S_e and S_i denote the power spectrum of the SAR echo and RFI, respectively. After transforming into the range–frequency domain via FFT, the subband division result is

$$F_t = \mathrm{FFT}(S_t) = F_{\mathrm{sub1}} + F_{\mathrm{sub2}} \tag{11}$$

where F_{sub1} and F_{sub2} represents the RFI-contaminated subband and the clean subband, respectively. Due to the holographic property of the subband spectrum [40], the two suimages can be obtained by applying the inverse FFT (IFFT)

$$\begin{cases} S_{\mathrm{sub1}} = \mathrm{IFFT}(F_{\mathrm{sub1}}) \\ S_{\mathrm{sub2}} = \mathrm{IFFT}(F_{\mathrm{sub2}}) \end{cases} \tag{12}$$

where S_{sub1} and S_{sub2} represent the RFI-contaminated subimage and clean subimage, respectively. These two power spectra are composed as follows:

$$\begin{cases} |S_{\mathrm{sub1}}|^2 \approx |S_{e1}|^2 + |S_i|^2 \\ |S_{\mathrm{sub2}}|^2 = |S_{e2}|^2 \end{cases} \tag{13}$$

where S_{e1} and S_{e2} represent the RFI-free power spectra, and S_i denotes the RFI power spectrum, respectively. After subtracting the power of the two subbands, the RFI artifact can be removed as follows

$$\begin{cases} |S_i|^2 \approx |S_{\mathrm{sub1}}|^2 - |S_{\mathrm{sub2}}|^2 \\ |S_e|^2 \approx |S_t|^2 - |S_i|^2 . \end{cases} \tag{14}$$

Since NBI occupies a relatively small bandwidth, the SSC method described above can achieve high-performance removal of NBI artifacts.

B. WBI Mitigation Based on SSC-SCDA

In 2022, an improved SSC method based on successive cancellation and data accumulation (SSC-SCDA) was proposed to mitigate WBI [42]. Initially, a segment of RFI-free power spectrum is selected as a clean subband for cancellation

$$|S_i|^2 \approx |\mathrm{Sub}_i|^2 - |\mathrm{Sub}_{\mathrm{clean}}|^2 \tag{15}$$

where $\mathrm{Sub}_{\mathrm{clean}}$ represent the initial clean subband, Sub_i denoted the adjacent RFI-contaminated subband, $|S_i|^2$ signifies the removed RFI, and the clean subband is updated as follows:

$$\begin{cases} |\mathrm{Sub}_{i_\mathrm{clean}}|^2 = |\mathrm{Sub}_i|^2 - |S_i|^2 \\ \mathrm{Sub}_{\mathrm{new_clean}} = \mathrm{Sub}_{\mathrm{clean}} + \mathrm{Sub}_{i_\mathrm{clean}}. \end{cases} \tag{16}$$

Then, the above operation is iterated to enable continuous cancellation and data accumulation of the clean subband. The mitigated result after separating all WBI artifacts is as follows:

$$|S_e|^2 = |S_t|^2 - |S_i|^2 \tag{17}$$

where S_t represents the RFI-contaminated data processing and S_i denotes the total WBI components.

C. Difference Ratio Operator

The difference ratio operator, a classic change detection technique, is employed to quantify feature changes between images [49], [50]. To highlight the local difference information induced by RFI artifacts, this study introduces this metric defined as follows:

$$D(x,y) = \frac{I_1(x,y) - I_2(x,y)}{I_1(x,y) + I_2(x,y)} \tag{18}$$

where x and y represent the position index of the corresponding pixel and I_1 and I_2 represent the RFI-contaminated image and the RFI-free image in the same scene, respectively.

IV. Methodology

In this section, a novel method for RFI artifact detection and mitigation is proposed as an enhanced version of RBSF. The scheme primarily consists of three steps: estimation of ground-truth amplitude based on ASSC, RFI artifact detection based on improved DRO (IDRO), and RFI artifact mitigation using RBSF through an adaptive dimension optimization strategy. The flowchart of the proposal is shown in Fig. 4, and its corresponding pseudocode is provided in Table I.

A. Ground-Truth Amplitude Estimation Based on ASSC

Considering the complex RFI types encountered in practical engineering applications, the proposed method integrates the original SSC with SSC-SCDA into an adaptive framework to estimate the amplitude of ground truth in the scene.

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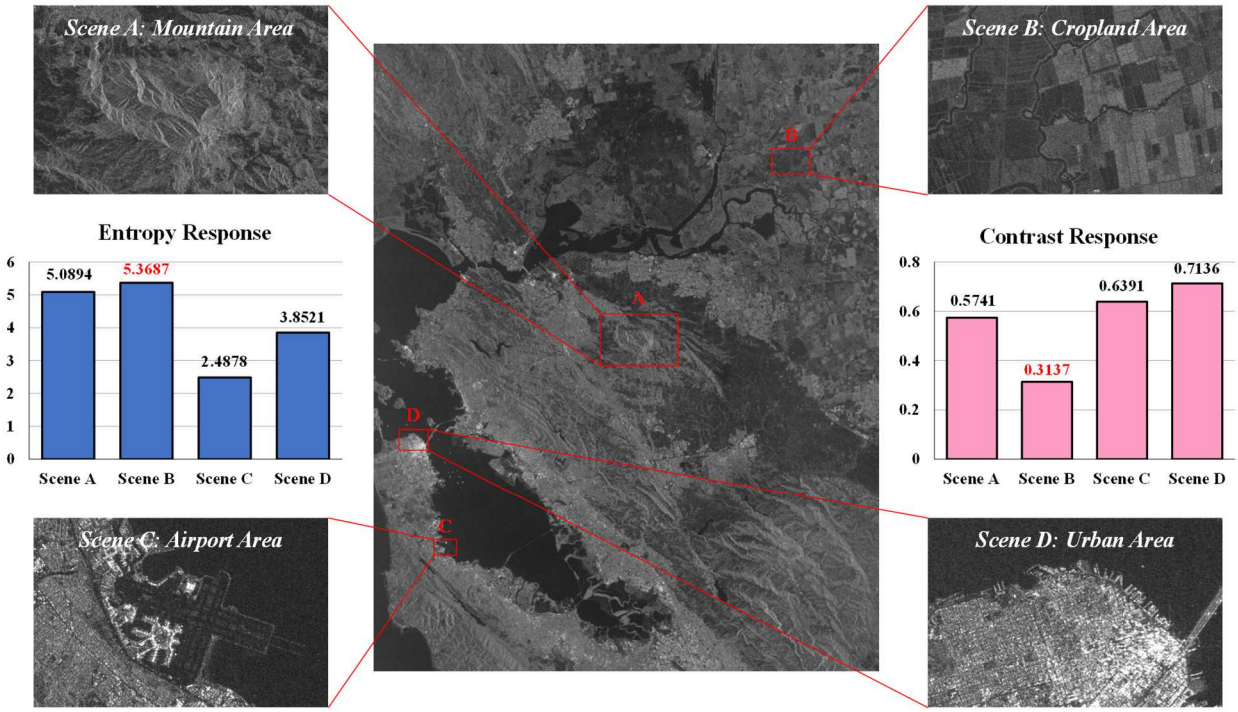


Fig. 3. Indicator response of the ground truth in different local regions is in a scene of an SAR image.

1) Range Spectrum Preprocess: Since phase compensation in the SAR imaging process may disrupt the spectral symmetry, leading to subband amplitude mismatch, the spectral amplitude is corrected via inverse phase compensation as follows:

$$S_p(\tau,\eta) = S_n(\tau,\eta) * e^{j\pi K\tau^2} \tag{19}$$

where τ and η represent the range time and azimuth time, K denotes the range modulation frequency, and S_n and S_p denote the focused image before and after correction, respectively.

2) Adaptive SSC: Compared with SAR echoes, RFI exhibits strong energy and 2-D spectrum coherence, leading to obvious abrupt variations in spectral domains. Therefore, the range-averaged spectrum is derived by applying FFT. Through kurtosis detection, RFIs are categorized into NBI and WBI based on bandwidth. As illustrated in Fig. 5, NBI is removed by directly canceling RFI-contaminated subbands, whereas WBI is separated through continuous cancellation of RFI-contaminated subbands and accumulation of clean subbands. To enable adaptive processing, the threshold condition "$B_m \gg B_{\mathrm{RFI}}$" is introduced to differentiate them. Thus, an approximately RFI-free result was obtained, where the ground-truth amplitude aligns more closely with the observed data.

B. RFI Artifact Detection Based on IDRO

Subsequently, under the premise that significant feature changes in the same scene are induced by RFI artifacts, an IDRO is introduced to achieve the detection and 2-D localization of RFI artifacts, which is defined as follows:

$$D(x,y) = \left| \frac{I_c(x,y) - |I_t(x,y)|}{I_c(x,y) + |I_t(x,y)| + \epsilon} \right| \tag{20}$$

where $I_c(x,y)$ and $I_t(x,y)$ denote the pixel values at the corresponding positions in the above estimation result and the RFI-contaminated image, respectively. ϵ serves as a smoothing factor to avoid division-by-zero errors. Notably, the pixel values in I_t are normalized to ensure data consistency. Subsequently, the detection result is further accentuated through binarization to obtain the RFI signature, which is defined as follows:

$$B(x,y) = \begin{cases} 1, & D(x,y) \geq k \\ 0, & D(x,y) < k \end{cases} \tag{21}$$

where k denote the segmentation threshold derived from the OTSU algorithm, and D represents the change detection result.

C. RFI Artifact Mitigation Based on Enhanced RBSF

As discussed in Section II, the optimization strategy proposed in [46] is suboptimal for complex types of RFI artifacts across diverse scenarios, and its mitigation efficiency is challenging to guarantee in practical applications. To address this limitation, this article introduces a novel optimization strategy to enhance the RBSF method during RFI mitigation, which flowchart is illustrated in Fig. 6.

1) Subblock Segmentation: Based on the RFI signature obtained by IDRO, RFI-contaminated regions are partitioned into subblocks. The block division strategy is enhanced by partitioning subblocks based on the row–column dimensions of both the amplitude estimation result and the RFI-contaminated image. In complex scenarios such as urban areas, where spatial fluctuations are significant, a finer subblock partitioning strategy is adopted in both azimuth and range dimensions.

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TABLE I

SPECIFIC PROCESS OF THE ASSC-RBSF ALGORITHM

Algorithm The ASSC-RBSF Algorithm

Inputs (Known Parameters):

$\boldsymbol{S}$	RFI-contaminated SLC image
$\boldsymbol{\lambda}$	Regularization parameter (>0)
$\boldsymbol{K}$	Range modulation frequency
$\boldsymbol{K_{max}}$	Max eigenvalue count
$\boldsymbol{k}$	OTSU segmentation threshold
$\boldsymbol{L}$	Subblock size
$\boldsymbol{\varepsilon}$	Smoothing factor to avoid division-by-zero

Internally Computed (Unknown Parameters):

$\boldsymbol{R_c}$	Clean amplitude estimate
$\boldsymbol{D}$	Difference ratio result
$\boldsymbol{B}$	Binary RFI mask
$\boldsymbol{K_u}$	Estimated subspace dimension
$\boldsymbol{U_r}$	RFI subspace eigenvectors
$\boldsymbol{I_u}$	Estimated RFI artifact
$\boldsymbol{R_e}$	Final RFI-mitigated image

Initialization:

Set $\boldsymbol{u=1}$; $\boldsymbol{R_c=0}$; $\boldsymbol{D=0}$; $\boldsymbol{B=0}$; $\boldsymbol{R_e=0}$;

Step 1: RFI Artifact Detection && 2-D Localization

Inputs: $\boldsymbol{S}$, $\boldsymbol{K}$

$S_p(\tau,\eta) = S(\tau,\eta) \times e^{j\pi K t^2}$

$F = FFT(S_p, range)$

For each i **in** F **do**

  K_i = Kurtosis(F_i)

  if $K_i >$ threshold then

    Mark F_i as RFI-subband

  else

    Mark F_i as clean-subband

  end if

end for

Apply SSC-SCDA:

  Cancel RFI subbands

  Accumulate clean subbands

$\boldsymbol{R_c} = IFFT(Clean\ spectrum, range)$

Outputs: $\boldsymbol{R_c}$

Step 2: Ground-truth Amplitude Estimation

Inputs: $\boldsymbol{R_c}$

For each $pixel\ (x,y)$ **in** $image$ **do**

  $D(x,y)$ = IDRO ($R_c(x,y), S(x,y)$)

end for

For each $pixel\ (x,y)$ **do**

  $B(x,y)$ = OTSU ($D(x,y), k$)

end for

Outputs: $\boldsymbol{D}, \boldsymbol{B}$

Step 3: Sub-block RFI Artifact Mitigation

Inputs: $\boldsymbol{S}, \boldsymbol{D}, \boldsymbol{B}, \boldsymbol{K_{max}}, \boldsymbol{L}$

$u = 1$

For each $subblock\ u$ **do**

  $S_u = subblock\ (S)$

  $R_u = U * \Lambda * U^H$

  $T_u = \max(\Lambda)$

  $K_u = number\ of\ \lambda_i\ where\ \lambda_i > T_u$

  $I_u = U_r * U_r^H * S_u$

  $S_{clean} = S_u - I_u$

Insert S_{clean} into R_e at position of subblock u

$u = u + 1$

end for

Outputs:

$\boldsymbol{R_e}$ RFI-mitigated image

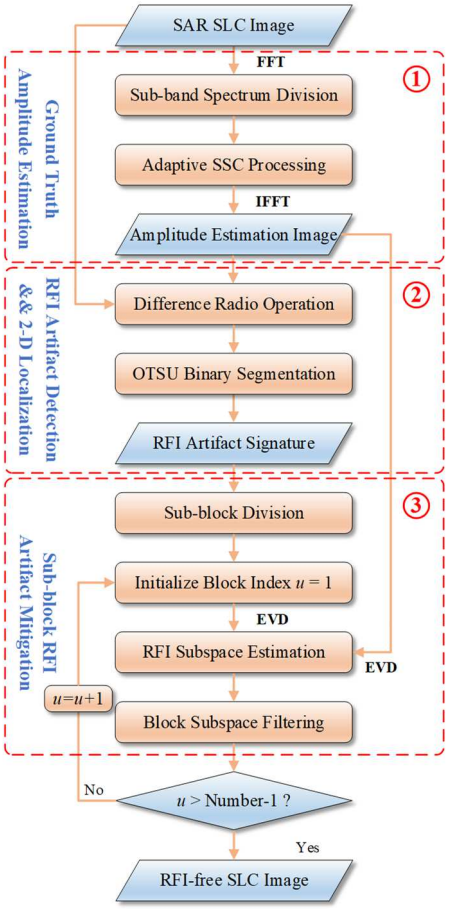


Fig. 4. Flowchart of the proposed approach.

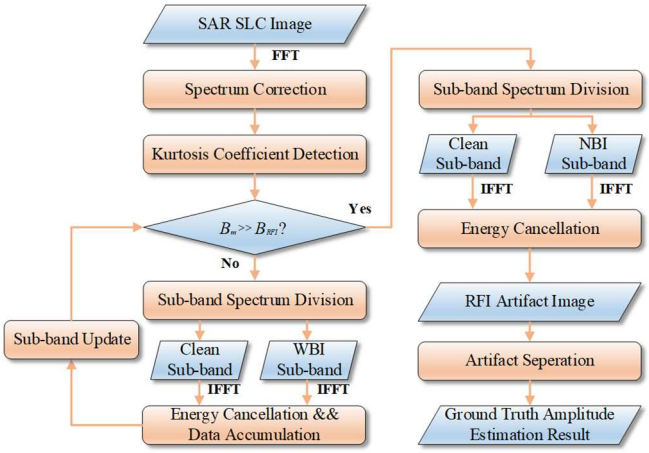


Fig. 5. Framework of the proposed ASSC processing.

This localized processing minimizes scene variation within each block and enhances the discrimination between RFI and background components.

2) Subblock RFI Subspace Dimension Estimation: For a target region with homogeneous RFI artifacts, the internal energy distribution may still fluctuate due to ground scattering points. Therefore, to further enhance the estimation accuracy of the RFI subspace dimension, the dimensions of all subblocks are estimated individually. For *i*th subblock in

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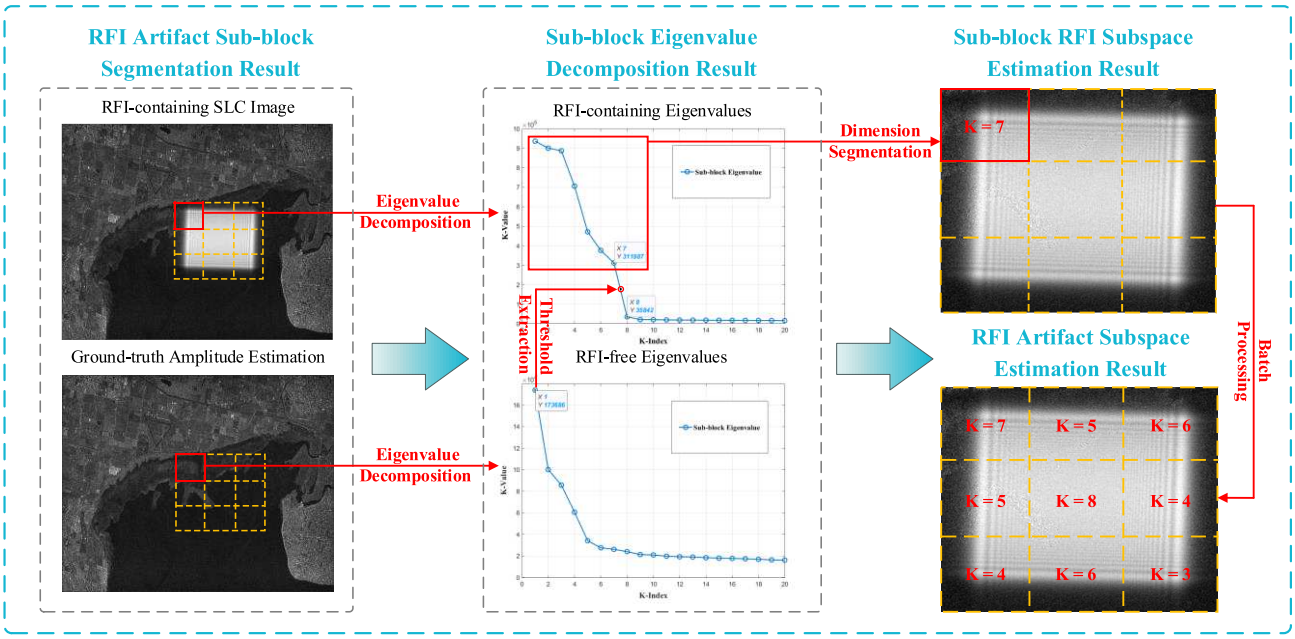


Fig. 6. Diagram of the proposed RFI subspace dimension optimization strategy.

the amplitude estimation result, the covariance matrix R_i is constructed and eigendecomposed as follows:

$$R_i = S_i S_i^T = \sum_{k=1}^{N} \lambda_k e_k e_k^{\mathrm{H}} \tag{22}$$

where S_i denote the numerical matrix, $\Sigma = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$ is the diagonal matrix containing all eigenvalues, and $U_i = [e_1, \ldots, e_N]$ are the eigenvectors corresponding to these eigenvalues. Based on the premise that the eigenvalue associated with the interference component is significantly larger, the dynamic segmentation threshold is set as follows:

$$t_{\mathrm{opt_}i} = \max(\lambda_1, \ldots, \lambda_N). \tag{23}$$

Iterate the above steps until the segmentation thresholds of all subblocks are obtained, thereby completing the subspace dimension estimation of the entire region contaminated by RFI artifacts. The dimensionality of the constructed subspace is adaptively chosen based on the scene complexity. In relatively homogeneous scenes such as farmlands, a lower dimension suffices. In contrast, for complex urban regions, the subspace dimension accounts for the higher similarity between RFI and background features, ensuring the robust separation.

3) RFI Artifact Reconstruction and Removal: For the corresponding regions in the RFI-contaminated SLC image, the identical subblock segmentation and eigendecomposition procedures are applied. Taking the ith subblock as an example, the optimal threshold $t_{\mathrm{opt_}i}$ is introduced to estimate the subspace dimension K_i of the RFI response, defined as follows:

$$K_i = \sum_{i=1}^{N} \mathbb{I}\left(\hat{\lambda}_i > t\right) \tag{24}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, which returns 1 if the condition holds and 0 otherwise. Then, the RFI subspace can be constructed using the eigenvectors corresponding to the K_i largest eigenvalues, which is defined as follows:

$$U_{i_r} = \mathrm{span}\{u_1, \ldots, u_{K_i}\}. \tag{25}$$

Afterward, the covariance matrix $\hat{R}_i$ is projected to obtain the RFI component

$$\hat{Q}_{i_r} = U_{i_r} U_{i_r}^H \hat{R}_i. \tag{26}$$

Finally, the corresponding RFI artifacts can be reconstructed and removed using the binary averaging method [40]. If the estimated dimension $K = 0$, it implies that no significant RFI component has been detected or separated, and thus, the RFI cancellation is deemed unsuccessful.

V. Experimental Results and Discussion

In this section, extensive simulated and measured experiments were conducted using C-band and L-band spaceborne SAR SLC data from Radarsat-1, Sentinel-1A, and ALOS-2 contaminated by RFI across diverse scenarios. The detection and mitigation performance of the proposal were quantified and discussed based on these datasets. Comparative tests were performed against the RBSF method, and the experimental results validate the effectiveness and superiority of the proposed approach.

A. Evaluation Indicators

The detection performance of the proposed method is quantitatively analyzed via simulated data. The RFI artifact in the image is manually annotated, with each label occupying 500 pixels. The evaluation index $F1$-score is introduced

$$F1 = \frac{2\mathrm{PR} \times \mathrm{RE}}{\mathrm{PR} + \mathrm{RE}} \tag{27}$$

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where PR denotes the detection precision, RE represent the recall rate, and their calculations are as follows:

$$\mathrm{PR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \times 100\% \tag{28}$$

$$\mathrm{RE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \times 100\%. \tag{29}$$

Specifically, true positive (TP) denotes a pixel correctly identified as RFI-contaminated, while false positive (FP) refers to a pixel incorrectly classified as containing RFI. Conversely, a false negative (FN) corresponds to an RFI-contaminated pixel misclassified as background. A higher $F1$-score generally signifies superior accuracy in THE RFI artifact detection.

The evaluation indicators for RFI mitigation are root-mean-square error (RMSE), average gradient (AG), and structural similarity index (SSIM). RMSE quantifies the performance by measuring the difference between images. For a target image sized $M * N$, the definition of RMSE is as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{S}\sum_{i=1}^{M}\sum_{j=1}^{N}\left(X_{i,j} - Y_{i,j}\right)^2} \tag{30}$$

where $S = M * N$ denotes the total number of pixels and i and j represent the pixel coordinates in the RFI-contaminated image X and RFI-free image Y, respectively. A smaller RMSE value usually indicates better mitigation performance.

AG is a widely adopted spatial-domain metric for evaluating image detail and texture clarity. It is defined as the average magnitude of local gradients in the horizontal and vertical directions across the entire image

$$\mathrm{AG} = \sum_{i=1}^{H-1}\sum_{j=1}^{W-1}\frac{\sqrt{\left(I_{i,j} - I_{i,j-1}\right)^2 + \left(I_{i-1,j} - I_{i,j}\right)^2}}{(H-1)(W-1)} \tag{31}$$

where i and j represent the pixel coordinates of image I. Generally, a higher AG indicates that the underlying structure of the image is enhanced after effective RFI mitigation.

The SSIM index is used to evaluate the similarity of ground-truth features before and after RFI artifacts mitigation in the target area, and its definition is as follows:

$$\mathrm{SSIM} = l(x,y) * c(x,y) * s(x,y) \tag{32}$$

where l, c, and s represent the brightness similarity coefficient, contrast similarity coefficient, and structural similarity coefficient, respectively, and x and y denote the two images before and after RFI artifacts mitigation.

For measured data, AG alone can be misled due to it may increase with residual striping. Adding equivalent number of looks (ENL), edge preservation (EP), and radiometric bias (RB) to achieve more accurate quantitative evaluation. ENL is used to directly measure the absolute improvement in equivalent visual acuity of RFI-containing areas after mitigation processing, which is defined as follows:

$$\mathrm{ENL}_{\mathrm{improved}} = \frac{\mu_a^2}{\sigma_a^2} - \frac{\mu_b^2}{\sigma_b^2} \tag{33}$$

where μ_a and σ_a represent the mean and standard deviation of the target area after RFI mitigation, respectively, while μ_b and σ_b are the corresponding values before processing.

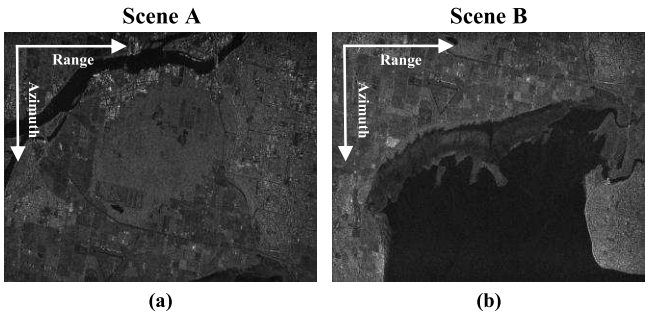


Fig. 7. Simulated experiments with (a) cropland area and (b) sea–land area.

TABLE II
MAIN PARAMETERS OF SIMULATED EXPERIMENTS

Parameters	Values
Carrier frequency	5.300 GHz
Sampling frequency	32.317 MHz
Efficient velocity	7062 m/s
Slant range	988.647 m
PRF	1256.981 Hz
Pulse width	41.74 μs
Pulse bandwidth	30.109 MHz
Carrier frequency of RFI	5.405 GHz
Bandwidth of RFI	1/3/9/16 MHz

EP is the core indicator for quantifying the RFI mitigation algorithm's ability to preserve the contrast of known terrain edges, defined as follows:

$$\mathrm{EP} = \frac{C_a}{C_b} \times 100\% \tag{34}$$

where C_b and C_a are the edge contrast before and after RFI mitigations respectively, and a higher EP represents better detail preservation effect after RFI mitigation.

RB is an indicator of the extent to which the image radiation characteristics are damaged by the quantitative image processing algorithm. It is used to evaluate the consistency between the image after RFI mitigation and the original image in the standard deviation of a uniform area, which is defined as follows:

$$\mathrm{RB} = \frac{|\sigma_a - \sigma_b|}{\sigma_b} \times 100\% \tag{35}$$

where σ_b and σ_a are the edge contrast before and after RFI mitigations, respectively, and a lower RB represents avoiding loss of more details in clean areas.

B. Experimental Result Based on Simulated SAR Data

1) Description of Simulated SAR Data: Compared with mountainous terrain and urban centers, oceans and croplands are characterized by lower contrast and higher image entropy, which features closely mimic RFI artifacts in SAR imagery, thereby degrading the effectiveness of subspace-based RFI optimization. To further validate the proposed method, two representative scenes from Radarsat-1 observation data are selected for case analysis in this section, as depicted in Fig. 7: cropland scene A and land-sea scene B. As shown in Table II, RFI artifacts with varying ISBRs and SIRs are

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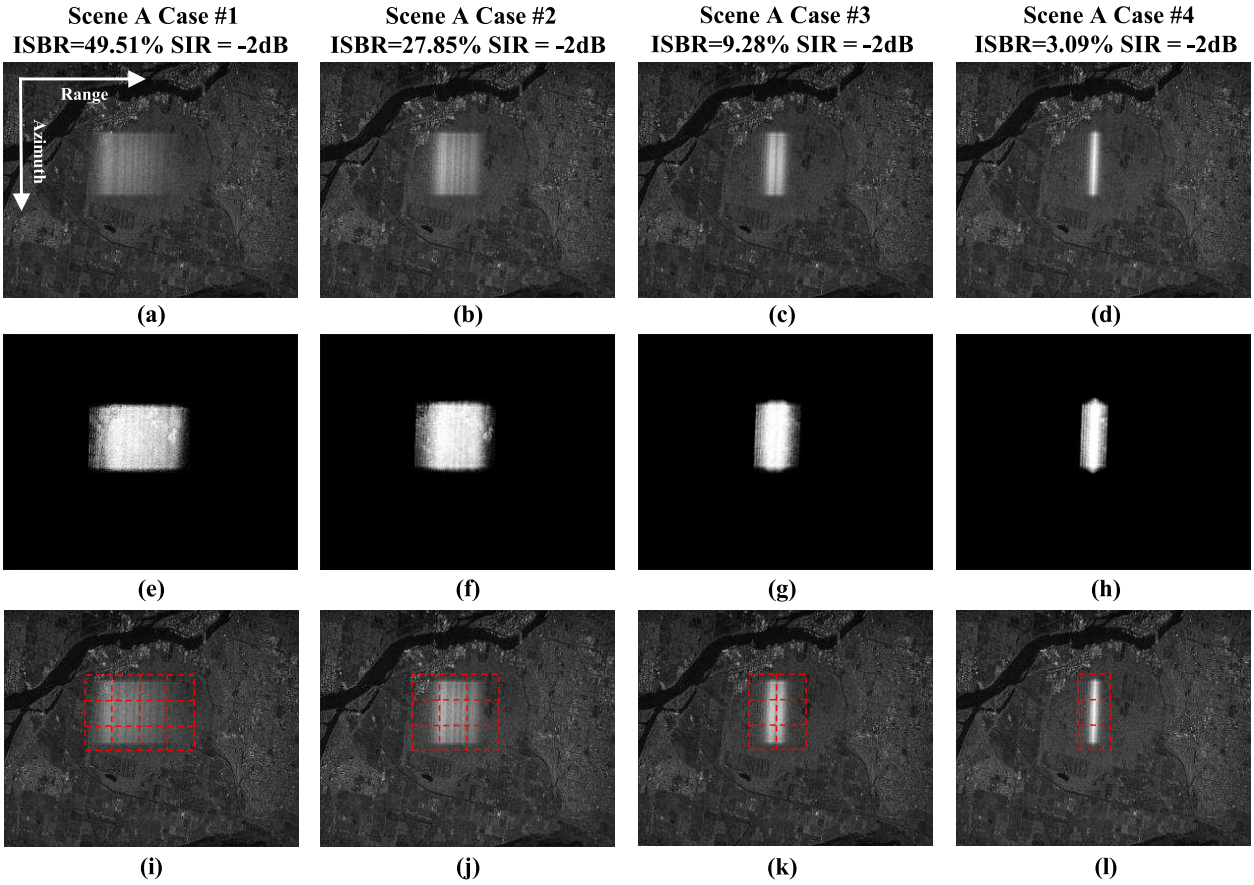


Fig. 8. RFI artifact detection and subblock segmentation result of Cases #1–4 in scene A. (a)–(d) RFI-contaminated cases under different ISBRs. (e)–(h) RFI signatures. (i)–(l) Sublock segmentation results.

TABLE III
*F*1-SCORES OF THE PROPOSED APPROACH UNDER DIFFERENT ISBRS

Experiment Cases		F1-score
Case #1	ISBR = 49.51% SIR = -2dB	0.9417
Case #2	ISBR = 27.85% SIR = -2dB	0.9532
Case #3	ISBR = 9.28% SIR = -2dB	0.8927
Case #4	ISBR = 3.09% SIR = -2dB	0.8831

separately added into the above-mentioned RFI-free SAR echo signals, which ensures representative RFI artifacts are covered in the target region after imaging processing. Although only farmland and near-shore scenes are presented in simulation experiments to highlight the limitations of the baseline RBSF method, additional terrain types, including forest and sea surface, were also tested, with results consistent in performance trends.

2) Experimental Result of Simulated Scene A: In scene A, RFI artifacts with different ISBRs are added to the same region under the same SIR (−3 dB). As illustrated in Fig. 8, as the ISBR decreases, the energy of the RFI artifact gradually concentrates, transitioning from the cloud-like artifact in Fig. 8(b) to the bright stripes in Fig. 8(d). Fig. 8(f)–(h) depicts the RFI detection results of different cases. As shown in Table III, the proposed method can effectively detect and locate artifacts in both two dimensions, with the detection accuracy exceeding 88%. However, due to the more dispersed energy of WBI artifacts along the range direction, the detection accuracy requires further improvement.

The RFI-contaminated areas are then divided into subblocks, and the mitigation outcomes are presented in Fig. 9. The first two columns display the mitigation results of these two methods. In the first row, both of them effectively separate WBI artifacts. In the subsequent three rows, as the ISBR decreases, the mitigation performance of the RBSF deteriorates, with more artifacts remaining in the target region of Case #4, which can be seen in the enlarged region of interest (ROI). In contrast, the proposed method separates the RFI artifacts more accurately across Cases #1–#4, as particularly evident in the magnified ROI shown in the last two columns. As tabulated in Table IV, while the RBSF method exhibits performance degradation processing with NBI artifacts, the proposed ASSC-RBSF method consistently maintains a lower RMSE of 0.2367, a higher AG of 3.3771, and an SSIM of 0.8917, demonstrating its superior performance.

To further analyze the performance degradation of RBSF in dealing with NBI artifacts, as shown in Fig. 10(a)–(d), the same local area across all cases was selected, and the variation pattern of the indicator response was recorded in Fig. 11. WBI typically occupies a broader frequency range,

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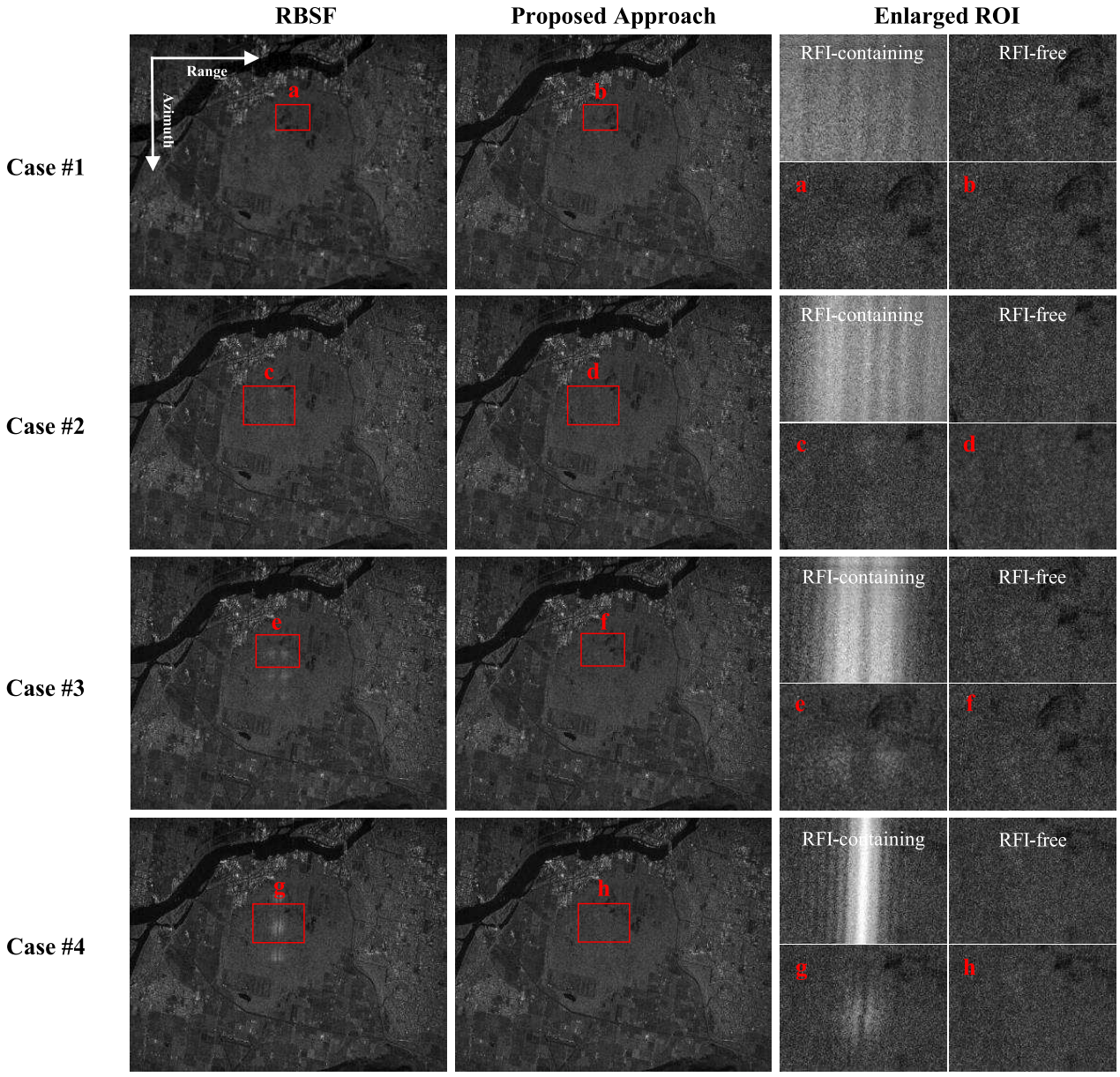


Fig. 9. RFI artifact mitigation result of Cases #1–4 in scene A through the RBSF and the proposed approach.

TABLE IV
RMSEs, AGs, AND SSIMs OF THE RBSF AND THE PROPOSED APPROACH WITH THE SIMULATED DATA UNDER DIFFERENT ISBRs

Experiment Cases \ Method		RMSE ↓		AG ↑		SSIM ↑	
		RBSF	Proposed Method	RBSF	Proposed Method	RBSF	Proposed Method
Case #1	ISBR = 49.51% SIR = -2dB	0.2631	**0.2472**	3.2495	**3.6124**	0.8972	**0.9131**
Case #2	ISBR = 27.85% SIR = -2dB	0.2771	**0.2315**	3.2426	**3.7131**	0.9014	**0.9326**
Case #3	ISBR = 9.28% SIR = -2dB	0.2836	**0.2389**	3.1368	**3.6425**	0.8253	**0.9207**
Case #4	ISBR = 3.09% SIR = -2dB	0.3655	**0.2367**	2.9204	**3.3771**	0.7862	**0.8917**

manifesting as large-area diffused artifacts in the image. This significantly increases the complexity and uncertainty of the image grayscale distribution, leading to a notable rise in image entropy. Meanwhile, WBI covers both high- and low-frequency components of the image, causing overall grayscale averaging and reducing contrast [51]. In contrast, NBI artifacts appear as periodic strip-like or linear artifacts, which reduce image entropy to some extent. Moreover, the grayscale jumps generated by the strip edges significantly enhance the contrast. As marked by the red circles in Fig. 10, the RBSF method clearly has insufficient dimension estimation in Cases #3 and #4. In contrast, the proposed ASSC-RBSF

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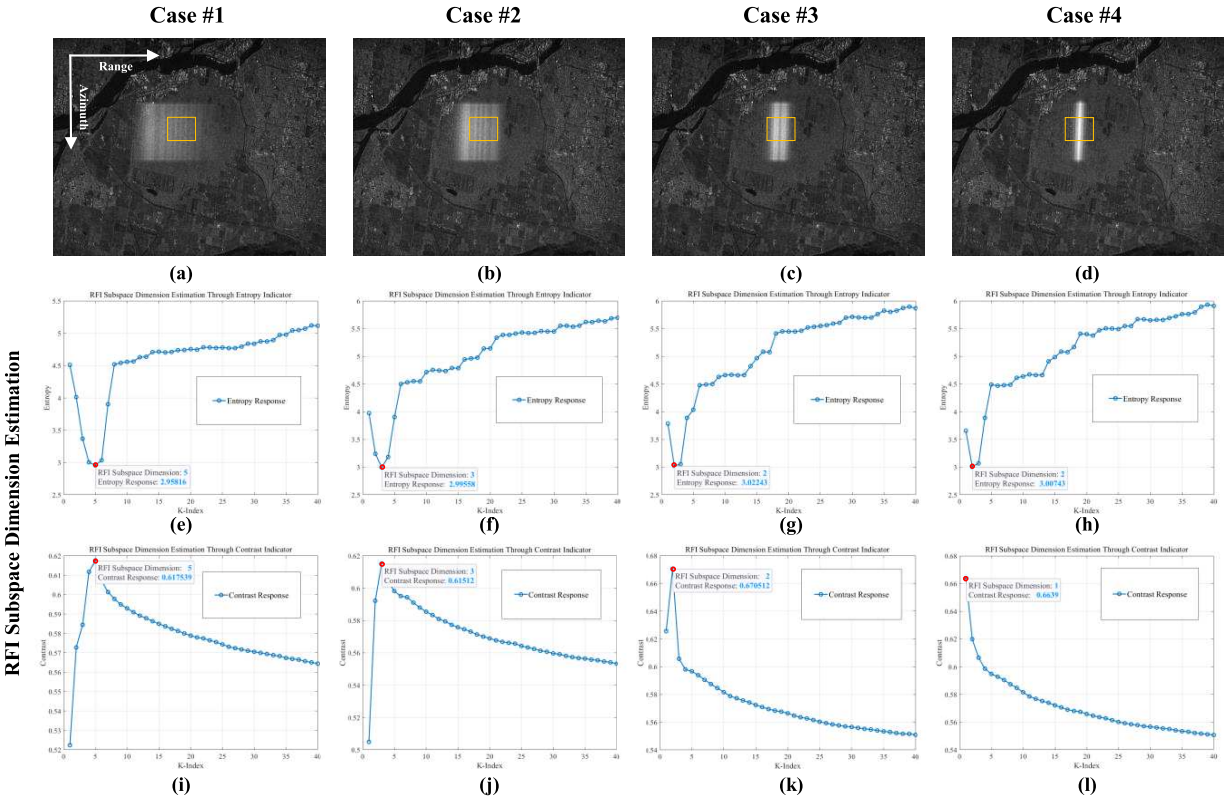


Fig. 10. RFI subspace dimension estimation through the RBSF method. (a)–(d) Same RFI-contaminated area in Cases #1–4. (e)–(h) Image entropy response curves. (i)–(l) Image contrast response curves.

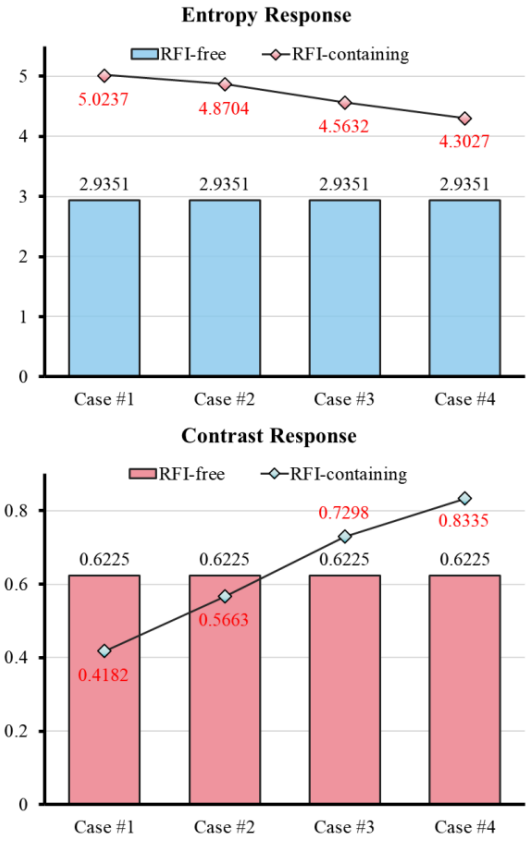


Fig. 11. Indicator response of target region in Cases #1–4 from scene A.

method yields optimal dimension of $K = 4$ and $K = 5$ for the respective cases. Based on the above analysis, this is attributed to the fact that the response patterns of NBI and the scene itself to the two indicators are more similar, thereby affecting the estimation accuracy. In addition, in some cases, the estimated dimensions corresponding to the two indicators are inconsistent, which may stem from the inherent characteristics of the cropland area causing indicator failure.

TABLE V

$F1$-SCORES OF THE PROPOSED APPROACH UNDER DIFFERENT SIRS

Experiment Cases		F1-score
Case #1	ISBR = 18.57% SIR = -5dB	0.9513
Case #2	ISBR = 18.57% SIR = -2dB	0.9334
Case #3	ISBR = 18.57% SIR = 2dB	0.9067
Case #4	ISBR = 18.57% SIR = 5dB	0.8871

3) Experimental Result of Simulated Scene B: In scene B, while maintaining the same bandwidth, RFI artifacts with varying SIRs were introduced into the land-sea region. As depicted in Fig. 12(a), as the SIR increases, the RFI energy gradually diminishes, transitioning from energy suppression in Case #1 to weak residual artifacts in Case #4. Fig. 12(f)–(h) illustrates the detection results for cases with different SIRs. Combining these results with the $F1$-scores in Table V, the proposed method effectively detects and localizes RFI artifacts and achieves a detection accuracy exceeding 88%. However, in high SIR conditions, some weak artifacts are overwhelmed by the strong background, leading to reduced accuracy to some extent.

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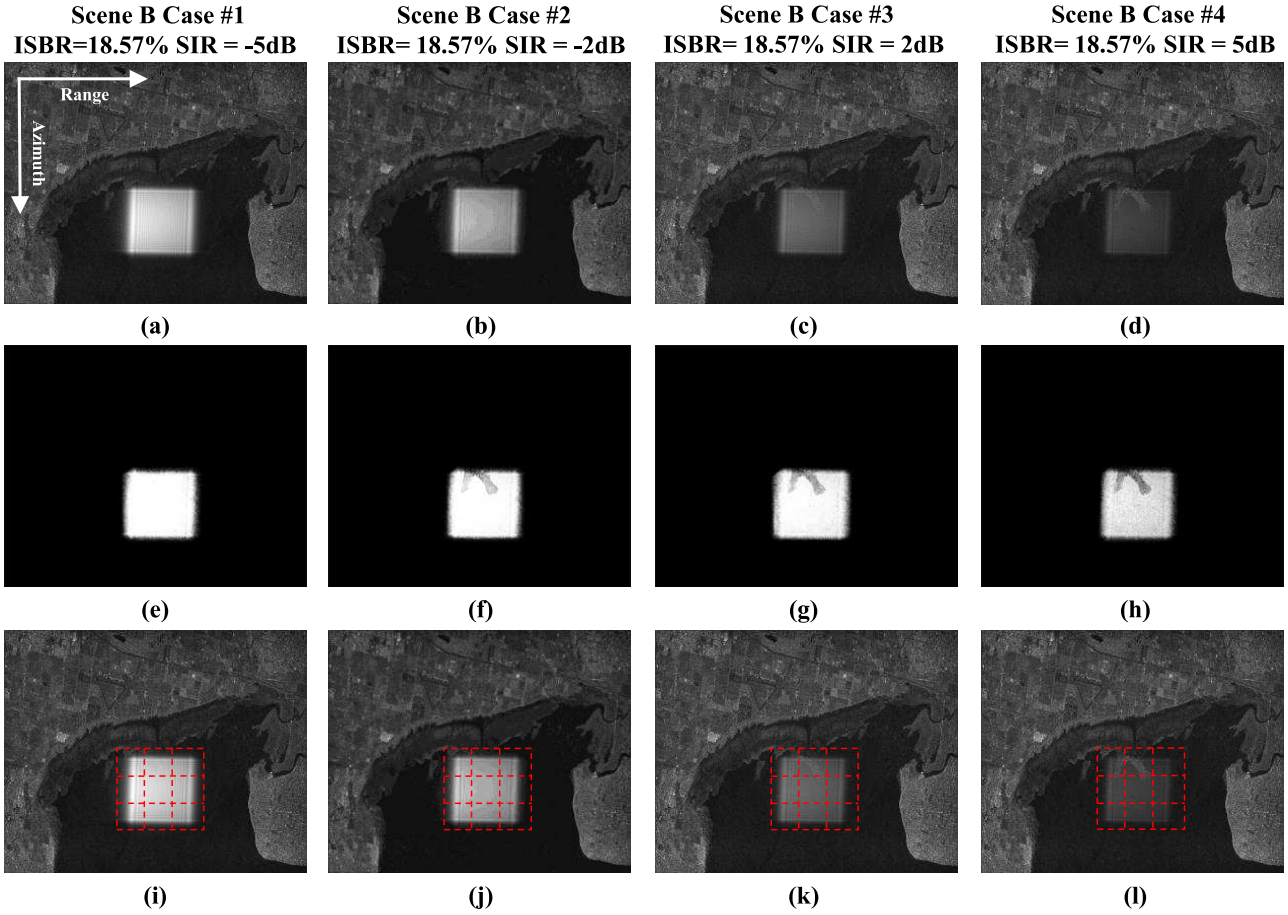


Fig. 12. RFI artifact detection and subblock segmentation result of Cases #1–4 in scene B. (a)–(d) RFI-contaminated cases under different SIRs. (e)–(h) RFI signatures. (i)–(l) Subblock segmentation results.

TABLE VI

RMSEs, AGs, AND SSIMs OF THE RBSF AND THE PROPOSED APPROACH WITH THE SIMULATED DATA UNDER DIFFERENT SIRs

Experiment Cases \ Method		RMSE ↓		AG ↑		SSIM ↑	
		RBSF	Proposed Method	RBSF	Proposed Method	RBSF	Proposed Method
Case #1	ISBR = 18.57% SIR = -5dB	0.2991	**0.2837**	1.2016	**1.3583**	0.8917	**0.9031**
Case #2	ISBR = 18.57% SIR = -2dB	0.2568	**0.2431**	1.7079	**1.8247**	0.9076	**0.9405**
Case #3	ISBR = 18.57% SIR = -2dB	0.2684	**0.2483**	1.3711	**1.7705**	0.8531	**0.9283**
Case #4	ISBR = 18.57% SIR = -5dB	0.3083	**0.2581**	1.1145	**1.5045**	0.8164	**0.9137**

The RFI-contaminated areas are then divided into subblocks, and the mitigation outcomes are presented in Fig. 13. The first two columns display the mitigation results of these two methods. The first two columns display the mitigation results of RFI with varying SIRs. In the first two rows, both methods effectively alleviate RFI artifacts. In the last two rows, as the SIR increases, the mitigation performance of the RBSF method deteriorates. In Case 4, more artifacts persist, as particularly evident in the enlarged ROIs shown in the last two columns. As shown in Table VI, the proposed method consistently maintains lower RMSE values and higher AG and SSIM values. However, it is observed that when the RFI energy is significantly strong (e.g., Case #1), the performance of both methods declines. This is attributed to the scenario where RFI components dominate the subspace, causing misjudgment and removal of background texture details, which in turn leads to local detail loss [26].

To further analyze the performance degradation of RBSF in mitigating RFI with low SIR, the same local region across all cases was selected, as shown in Fig. 14(a)–(d), and the variation pattern of the indicator response was shown in Fig. 15. When the SIR is low, the grayscale distribution in the local region tends to homogenize, with original texture details severely obliterated, leading to a notable increase in image entropy. Concurrently, contrast decreases significantly due to the attenuation of grayscale jumps [52]. Conversely, grayscale disturbances in the local region are relatively minor in high SIR conditions, causing only a slight increase in image entropy

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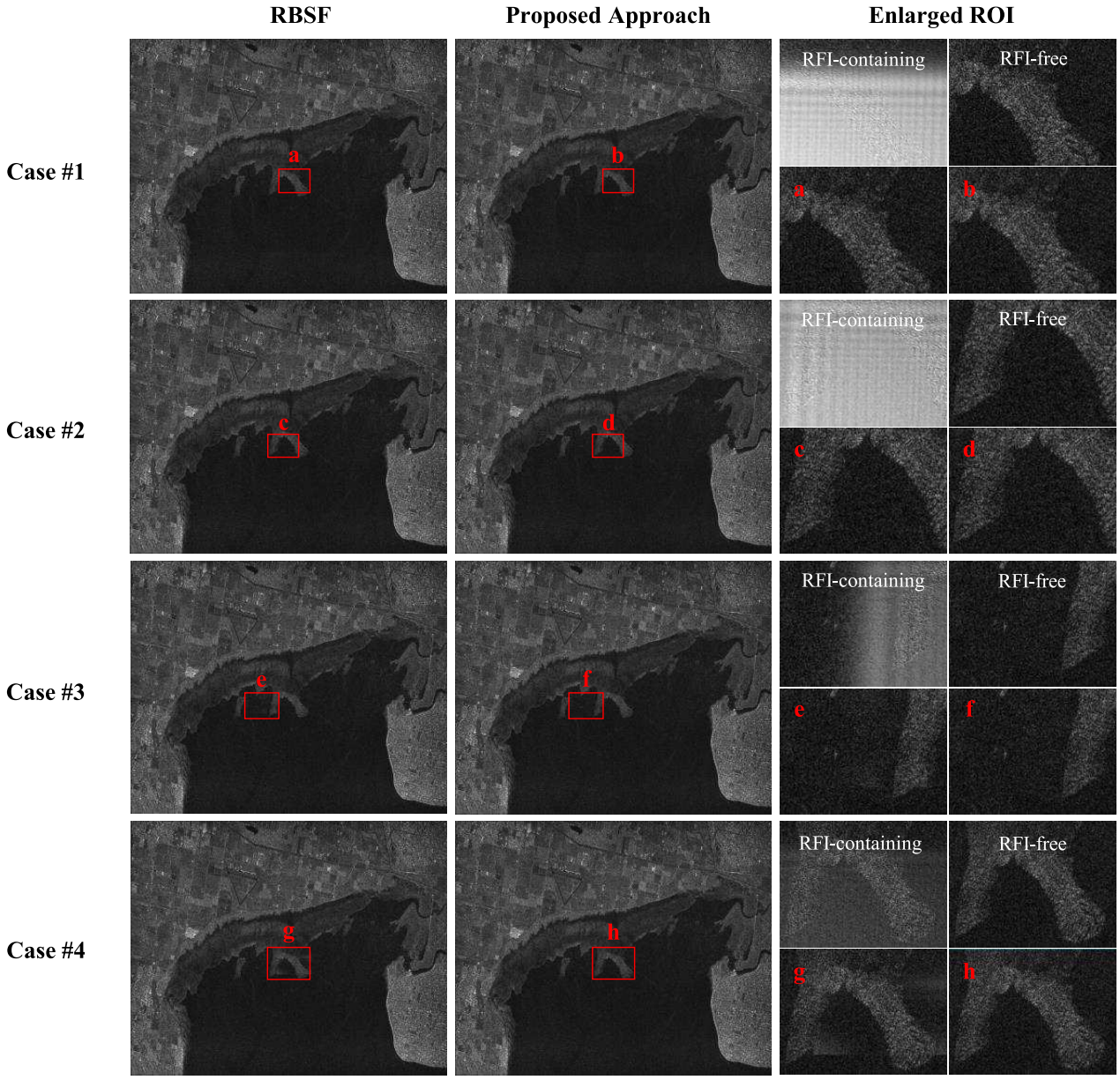


Fig. 13. RFI artifact mitigation result of Cases #1–4 in scene B through the RBSF and the proposed approach.

and a modest reduction in contrast [53]. As marked by the red circles in Fig. 14, the RBSF method clearly has insufficient dimension estimation in Cases #3 and 4. Based on the above analysis, this is attributed to the fact that the response patterns of high-SIR RFI are more similar to the scene itself, thereby degrading the estimation accuracy. In addition, in certain cases, the estimated dimensions derived from the two indicators are inconsistent, which may stem from the inherent characteristics of the sea–land area that cause indicator failure.

C. Experimental Result Based on Measured SAR Data

1) Description of Measured SAR Data: In this section, the effectiveness and superiority of the proposed ASSC-RBSF method are further validated through experiments. These experiments utilize spaceborne SAR level-1 products from the C-band satellite Sentinel-1A (TOPS mode) and L-band satellite ALOS-2 (ScanSAR mode). As shown in Figs. 16 and 17, the measured data were acquired from real observations in representative regions, including urban areas with dense structures and strong multipath effects, sea–land interfaces with backscatter discontinuity, mountains with obvious fluctuation and open-sea surfaces. Before RFI mitigation, a series of preprocessing steps including orbit correction, radiation correction, and deburst has been completed.

2) Experimental Result Based on Sentinel-1A SAR Data #1: Fig. 18 shows a scene of VH-polarization SAR data acquired near Hafar Al-Batin, Saudi Arabia, on March 31, 2021, which scenario spans 1569 $\times$ 6224 pixels. The block-shaped and strip-shaped RFI artifacts with varying energy levels are distributed along the range direction over built-up areas and desert regions. The latter two rows in Fig. 18 displays the RFI signature corresponding to the detection result of the above RFI-contaminated data through the lambda-1 detector [9] and

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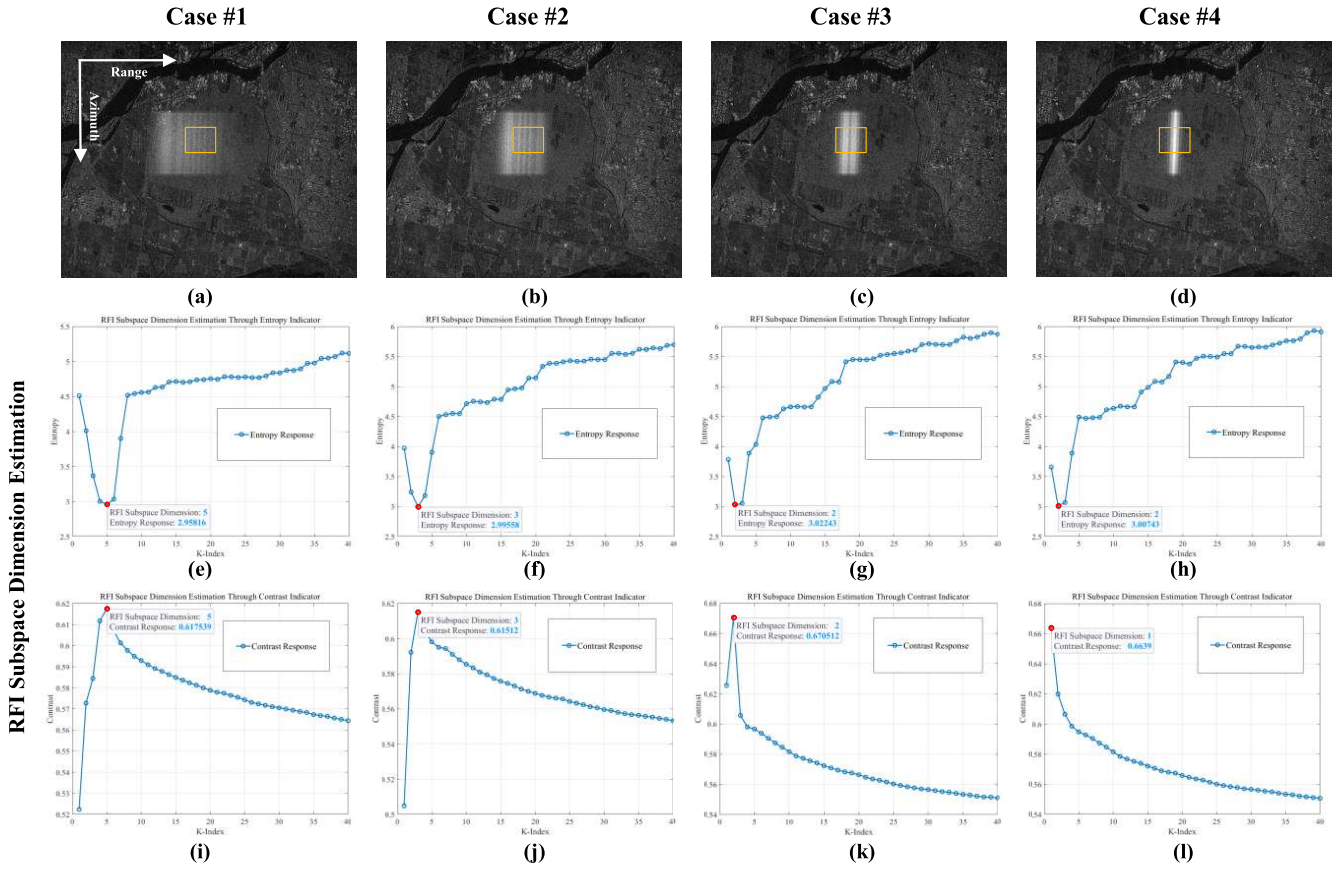


Fig. 14. RFI subspace dimension estimation through the RBSF method. (a)–(d) Same RFI-contaminated area in Cases #1–4. (e)–(h) Image entropy response curves. (i)–(l) Image contrast response curves.

proposed method. As reflected in the enlarged ROIs, compared with Lambda-1 detector that performs RFI detection on a blockwise basis, the proposed method achieves RFI artifacts detection and 2-D spatial localization at the pixel level, both of which serve as a robust reference for the subsequent block filtering procedure.

Based on the identified RFI signatures, subblock division and dynamic threshold segmentation are then performed into filter RFI artifacts in the contaminated regions. The first column of Fig. 19 presents the mitigation results of RBSF, RPCA, and the proposed method, respectively. In the enlarged areas, residual RFI artifacts are visible in the desert region after RBSF processing. In contrast, the proposed method more accurately removes the RFI components while better preserving background details. As shown in Table VII, the proposed method achieves better indicator responses from the quantitative evaluation results, demonstrating its improvement in RFI mitigation performance.

3) Experimental Result Based on Sentinel-1A SAR Data #2: Fig. 20 shows a scene of VV-polarized SAR data acquired on March 9, 2022, in the surrounding areas of Shanghai, China, which scenario spans 3441 × 12 970 pixels. Numerous overlapping RFI artifacts are revealed distributed along the range direction across seaport and offshore zones in the imagery. Meanwhile, the latter two rows in Fig. 20 displays the RFI signature corresponding to the measured data. As reflected in the enlarged ROIs, compared with the Lambda-1 detector that performs RFI detection on a blockwise basis, the proposed method achieves RFI artifacts detection and 2-D spatial localization at the pixel level, both of which serve as a robust reference for the subsequent block filtering procedure.

Based on the above RFI signature, the subblock division and dynamic threshold segmentation are then performed to filter RFI artifacts in the contaminated regions. The first column in Fig. 21 presents the mitigation results of RBSF, RPCA, and the proposed method, respectively. As reflected in the last two columns, residual RFI artifacts are visible in the seaport region after RBSF processing. In contrast, the proposed method more accurately removes the RFI components while better preserving background details. As shown in Table VII, the proposed method achieves better indicator responses from the quantitative evaluation results, demonstrating its improvement in RFI mitigation performance.

4) Experimental Result Based on Sentinel-1A SAR Data #3: Fig. 22 shows a scene of VH-polarized SAR data acquired on September 2, 2021, in the surrounding areas of Kaesong, North Korea, which scenario spans 780 × 3144 pixels. Several RFI artifacts are revealed distributed along the range direction across the mountain and valley zones in the imagery. Meanwhile, the latter two rows in Fig. 22 displays the RFI signature corresponding to the measured data. As reflected in the enlarged ROIs, compared with Lambda-1 detector that performs RFI detection on a blockwise basis, the proposed method achieves RFI artifacts detection and 2-D spatial local-

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TABLE VII
AGs, ENLs, EPs, AND RBs WITH THE MEASURED DATA

Target Scenes	AG ↑			ENL ↑			EP ↑			RB ↓		
	RBSF	RPCA	Proposed Method	RBSF	RPCA	Proposed Method	RBSF	RPCA	Proposed Method	RBSF	RPCA	Proposed Method
Al-Batin	0.6466	0.6537	**0.6801**	2.95	2.78	**3.71**	81.67%	88.63%	**91.37%**	13.61%	14.18%	**11.36%**
Shanghai	0.5281	0.5013	**0.5414**	6.51	4.37	**9.35**	79.49%	76.31%	**93.56%**	18.63%	23.67%	**10.14%**
Kaesong	0.2204	0.2491	**0.2869**	0.81	1.29	**2.14**	83.64%	79.29%	**90.37%**	22.61%	17.99%	**12.93%**
Doha	0.3671	0.3409	**0.3913**	2.48	3.61	**4.91**	81.77%	85.67%	**88.65%**	19.91%	23.49%	**12.47%**

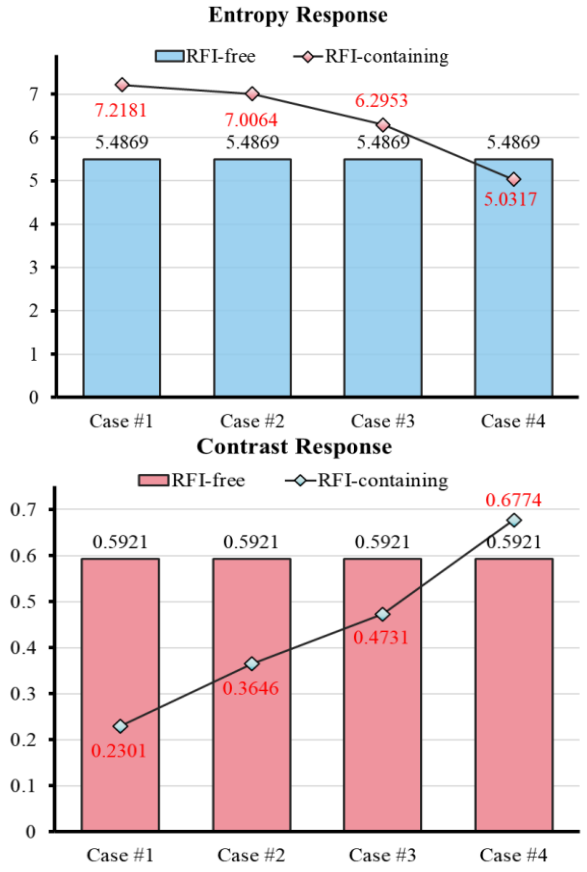


Fig. 15. Indicator response of target area in Cases #1–4 from scene B.

ization at the pixel level, both of which serve as a robust reference for the subsequent block filtering procedure.

Based on the above RFI signature, subblock division and dynamic threshold segmentation are then performed to filter RFI artifacts in the contaminated regions. The first column in Fig. 23 presents the mitigation results of RBSF, RPCA, and the proposed method, respectively. As reflected in the last two columns, residual RFI artifacts are visible in the seaport region after RBSF processing. In contrast, the proposed method more accurately removes the RFI components while better preserving background details. As shown in Table VII, the proposed method achieves better indicator responses from the quantitative evaluation results, demonstrating its improvement in the RFI mitigation performance.

5) Experimental Result Based on ALOS-2 SAR Data: Fig. 24 shows a scene of HH-polarized SAR data acquired from ALOS-2 PALSAR-2 on September 16, 2019, in the surrounding areas of Doha, Qatar, which scenario spans 1240 × 5752 pixels. In comparison to C-band SAR satellites operating at higher frequencies (4–8 GHz), L-band SAR satellites (1–2 GHz) occupy a lower frequency spectrum shared by numerous civilian and military communication systems, rendering them more vulnerable to complex RFI due to intensified spectral congestion and diverse interference sources [54]. Complex types of RFI artifacts cover the offshore areas in the above image, which are widely distributed in both the range and azimuth directions. The latter two rows in Fig. 24 displays the RFI signature corresponding to the measured data. As reflected in the enlarged ROIs, compared with the Lambda-1 detector that performs RFI detection on a blockwise basis, the proposed method achieves RFI artifacts detection and 2-D spatial localization at the pixel level, both of which serve as a robust reference for the subsequent block filtering procedure.

Thus, the subblock division and dynamic threshold segmentation are then performed to filter RFI artifacts in the contaminated regions. The first column in Fig. 25 presents the mitigation results of RBSF, RPCA, and the proposed method, respectively. As reflected in the last two columns, residual RFI artifacts are visible in the seaport region after RBSF processing. In contrast, the proposed method more accurately removes the RFI components while better preserving background details. As shown in Table VII, the proposed method achieves better indicator responses from the quantitative evaluation results, demonstrating its improvement in the RFI mitigation performance. It is noteworthy that the ScanSAR mode data acquired by ALOS-2 PALSAR-2 are processed using a full-aperture algorithm for 2-D energy focusing, which may introduce blurring and geometric distortion of dynamic targets within the scene (e.g., seaside ships). Consequently, the detection and mitigation performance of the proposed method for RFI artifacts is inherently constrained to a certain extent.

D. Discussion on Subresolution Targets Preservation and Computational Complexity

To quantitatively evaluate the capability of the proposed method in preserving subresolution targets under low-SCR conditions, we introduce the target energy retention rate

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Fig. 16. Measured Sentinel-1A SLC data from surrounding areas of (a) Al-Batin, (b) Shanghai, and (c) Kaesong.

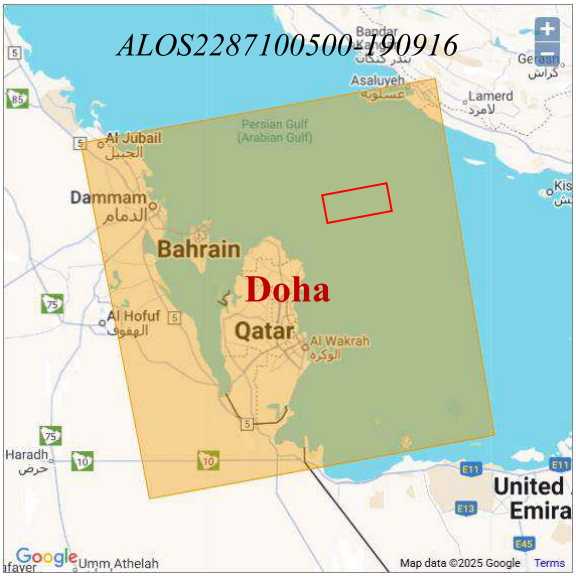


Fig. 17. Measured ALOS-2 SLC data from surrounding areas of Doha.

TABLE VIII
TERR OF THE RBSF AND THE PROPOSED APPROACH

Measured Scenes \ Methods	RBSF	Proposed Method
Shanghai, China (S1-A)	78.2%	**86.51%**

(TERR) as an evaluation metric, which is defined as follows:

$$\mathrm{TERR} = \frac{E_{\mathrm{RFI}}}{E_m} \times 100\% \tag{36}$$

where E_{RFI} denotes the total intensity (sum of pixel values) in a manually selected region containing the weak target in the original RFI-contaminated image and E_m refers to the corresponding intensity in the processed image after RFI mitigation.

The results, as summarized in Table VIII, clearly demonstrate that the proposed method achieves a higher TERR value, indicating superior preservation of subresolution targets compared with the original RBSF. This indirectly reflects the method's potential to support downstream tasks such as change detection and object monitoring. Moreover, the proposed method enhances the spatial integrity and amplitude consistency of the rconstructed image, which are essential prerequisites for interferometric processing and classification tasks.

TABLE IX
RUNNING TIMES OF THE RBSF AND THE PROPOSED APPROACH

Measured Scenes \ Methods	RBSF	Proposed Method
Al-Batin, Saudi Arabia (S1-A)	23.8248	**7.8959**
Shanghai, China (S1-A)	15.9217	**4.1826**
Kaesong, North Korea (S1-A)	11.3867	**2.0981**
Doha, Qatar (ALOS-2)	14.1224	**3.9301**

The primary time cost of the RBSF method originates from the optimization process of the interference subspace dimension parameter K, which is estimated by integrating image entropy and contrast metrics. For each candidate K, the local covariance matrix must undergo eigenvalue decomposition, interference components must be reconstructed, and image quality metrics must be calculated. For an SLC image of size $M * N$, segmented into subblocks of size $L * L$ with the total number of subblocks $N_b = MN/L^2$, the computational complexity of the algorithm is

$$C_{\mathrm{RBSF}} = \mathcal{O}\left(N_b K_{\max} L^3\right) = \mathcal{O}\left(MNK_{\max} L\right) \tag{37}$$

where $K_{\max}$ denotes the maximum subspace dimension requiring estimation, which is set to 40 for both simulated and measured data in this article.

The proposed ASSC-RBSF method exhibits a broader range of applicable conditions and demonstrates effective mitigation of complex RFI types across diverse scenarios. Its primary time overhead arises from two key processing steps: the adaptive subband cancellation and the extraction of corresponding dynamic segmentation thresholds. The computational complexity of the algorithm can be expressed as follows:

$$\begin{aligned} C_{\text{ASSC-RBSF}} &= C_{\mathrm{ASSC}} + C_{\mathrm{RBSF}} \\ &= \mathcal{O}\left(MN \log MN\right) + \mathcal{O}\left(MN\right) \\ &\approx \mathcal{O}\left(MN \log MN\right). \end{aligned} \tag{38}$$

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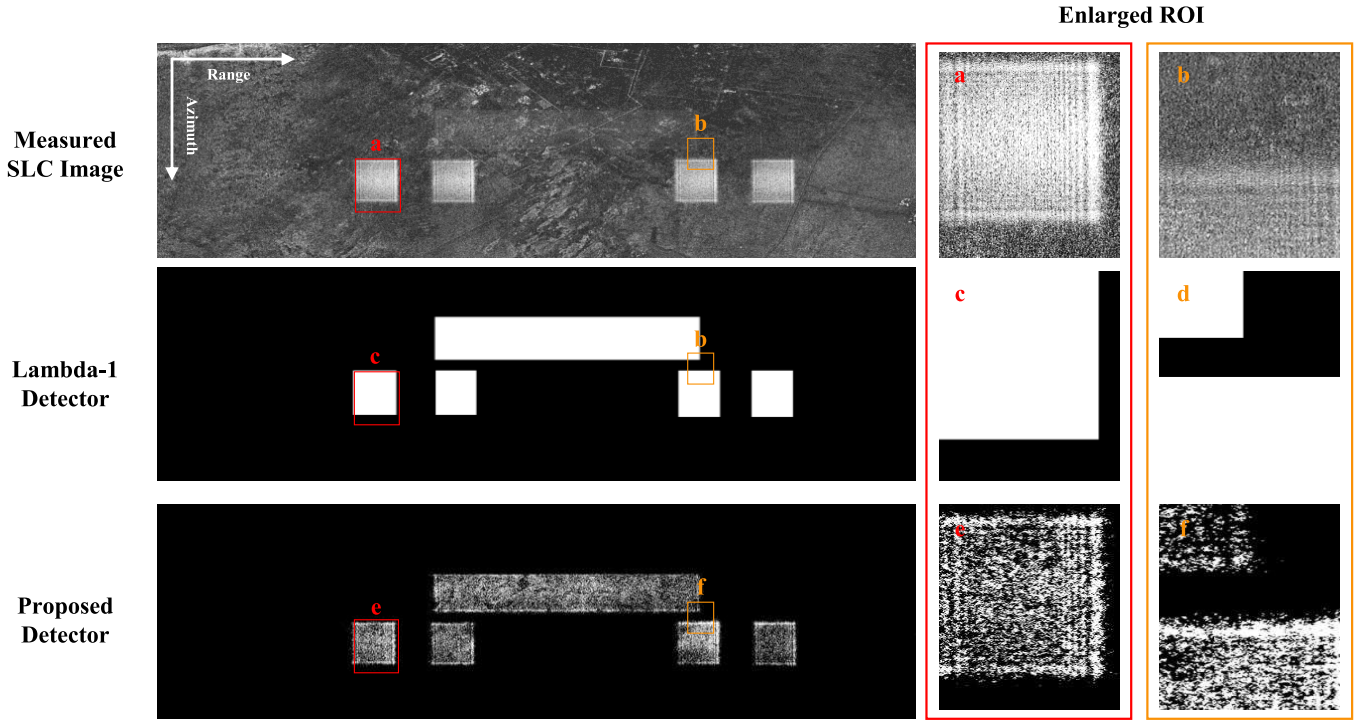


Fig. 18. Detection result of RFI artifacts based on Lambda-1 and the proposed detector in the AI-Batin surrounding area acquired on March 31, 2021.

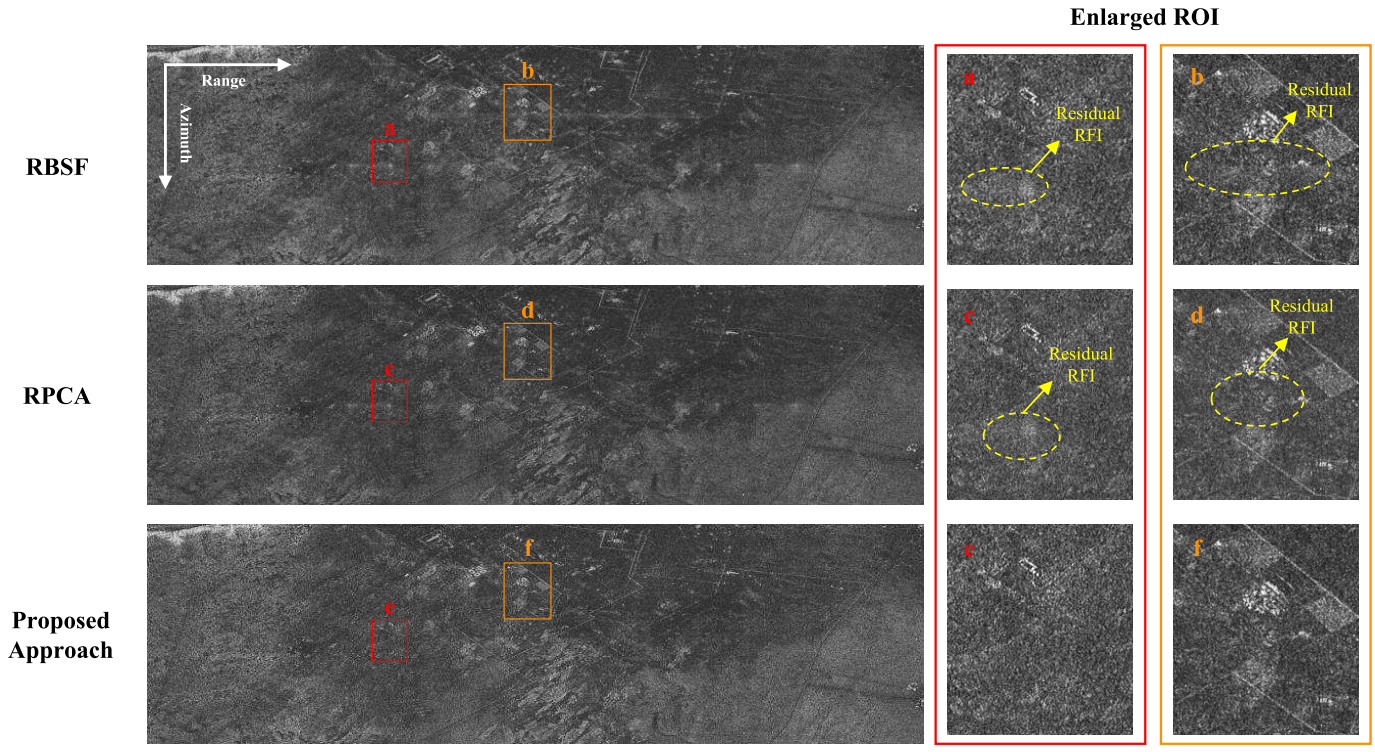


Fig. 19. RFI artifact mitigation result based on the RBSF, RPCA, and proposed approach.

In practical engineering applications, where SAR images often exceed 10^6 pixels, ASSC-RBSF's logarithmic scaling with image size ensures faster processing, making it more suitable for real-time and large-batch RFI mitigation tasks. To verify the computational efficiency, this study conducted experiments and recordings using measured spaceborne SAR data. All programs were implemented in MATLAB R2018a and tested on a computer configured with an Intel Xeon Gold

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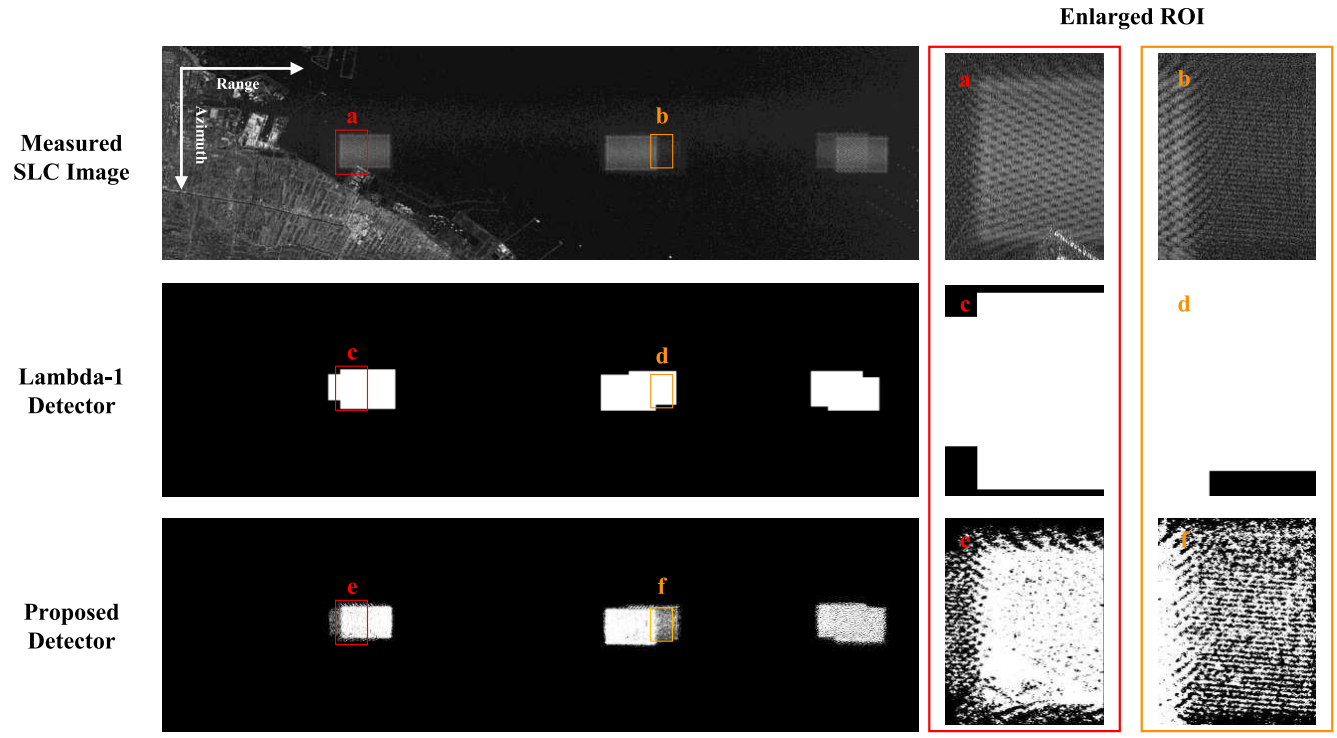


Fig. 20. Detection result of RFI artifacts based on Lambda-1 and proposed detector in Shanghai surrounding area acquired on March 9, 2022.

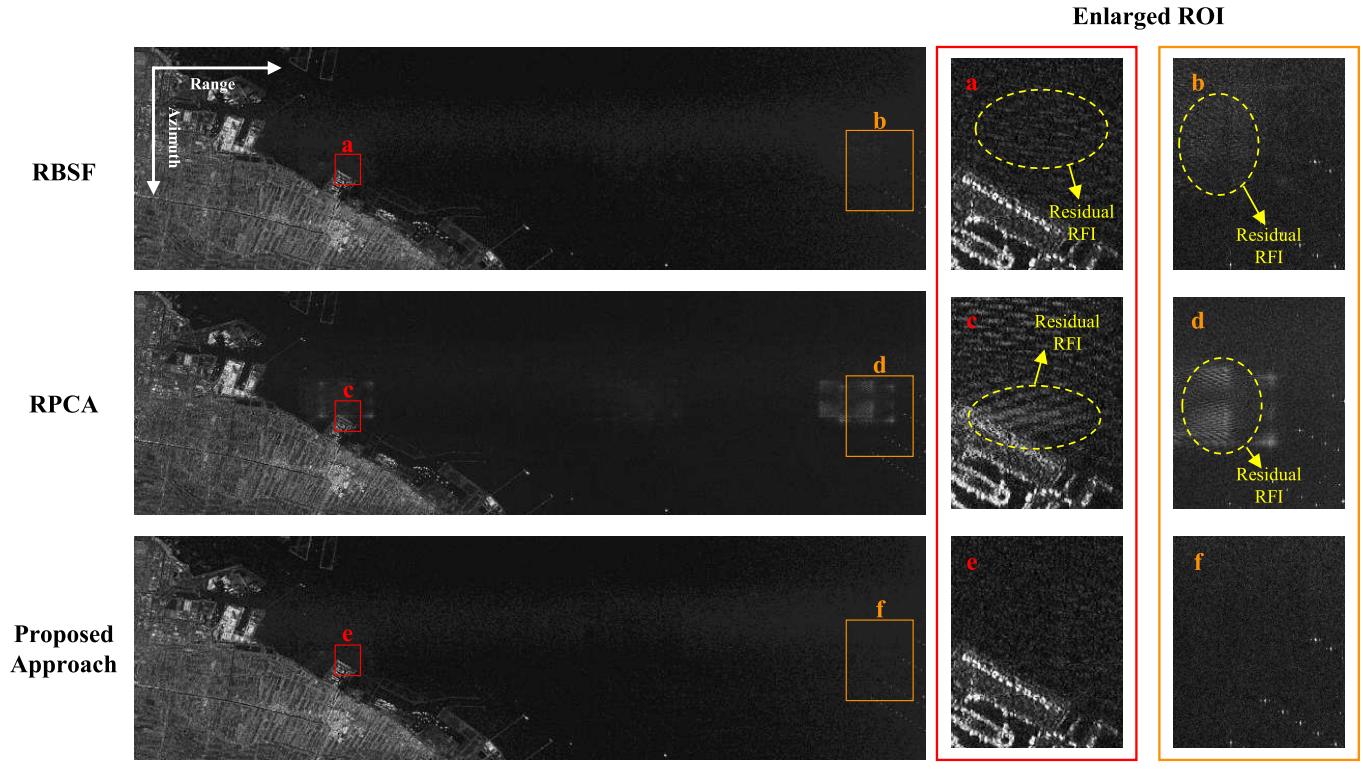


Fig. 21. RFI artifact mitigation result based on the RBSF, RPCA, and proposed approach.

6248R processor (3.00 GHz) and 24 GB of memory. As shown in Table IX, the proposed method significantly reduces the running time compared to the original RBSF method, primarily by avoiding the optimization problem-solving process for subspace dimension estimation.

VI. Conclusion

In recent years, the increasing proliferation of communication devices with overlapping operating frequency bands has led to more frequent and severe impacts of RFI on SAR systems. This article proposes a novel method framework,

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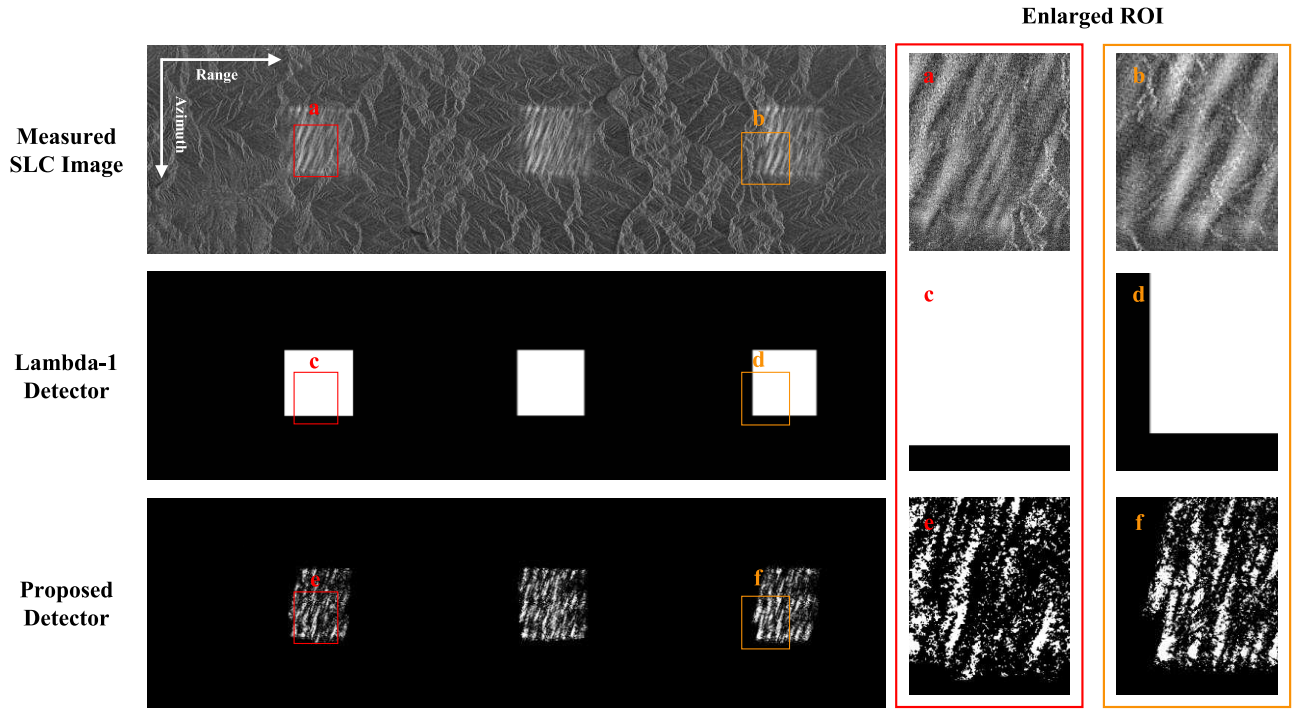


Fig. 22. Detection result of RFI artifacts based on Lambda-1 and the proposed detector in the Kaesong surrounding area acquired on September 2, 2021.

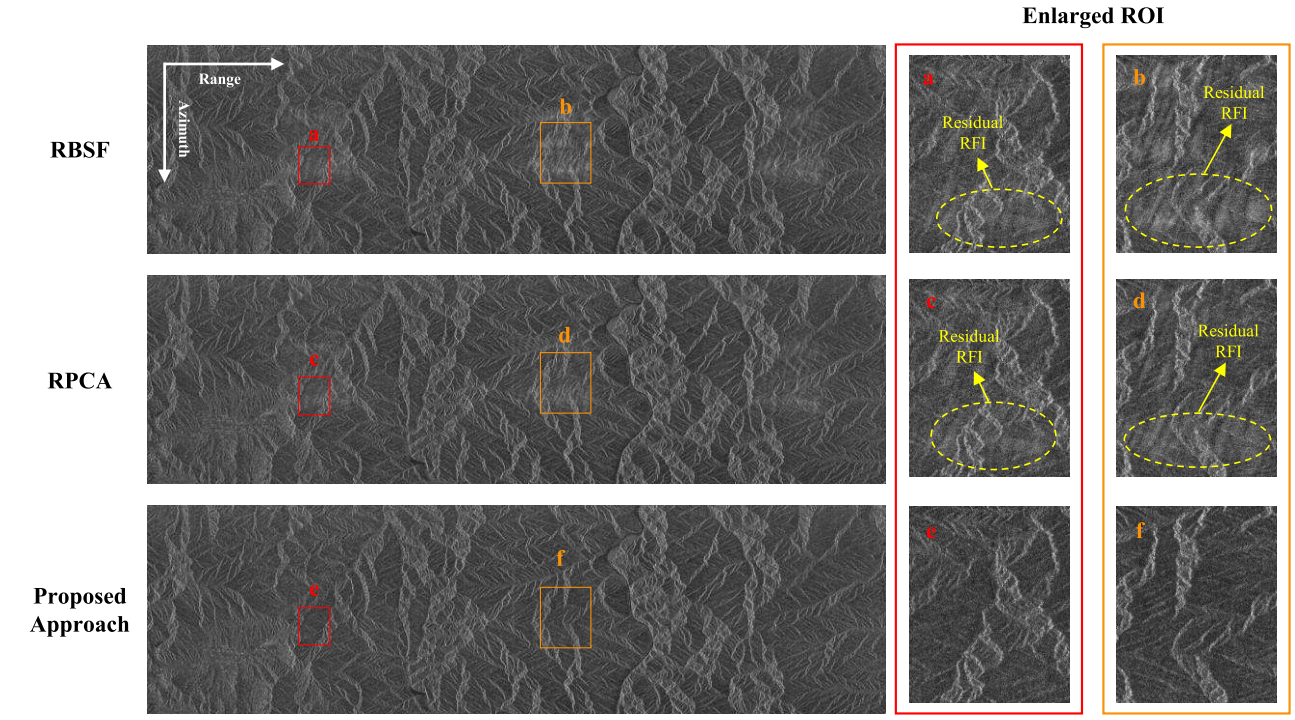


Fig. 23. RFI artifact mitigation result based on the RBSF, RPCA, and proposed approach.

namely, ASSC-RBSF methods for mitigating RFI artifacts in SLC SAR images. When the estimation result of the ground-truth amplitude is obtained, ASSC-RBSF employs the change detection technology to achieve high-precision RFI detection and artifact localization in both two dimensions. Based on these results, the method enables accurate estimation of the RFI subspace dimension, effectively addressing the performance degradation issue of the original RBSF method dealing with complex RFI types and diverse scenarios.

Experiments were performed using simulated and measured SLC data acquired from multiple spaceborne SAR satellites. The results demonstrate that the proposed method serves as an enhanced version of the original RBSF in mitigating complex types of RFI artifacts across diverse scenarios, which

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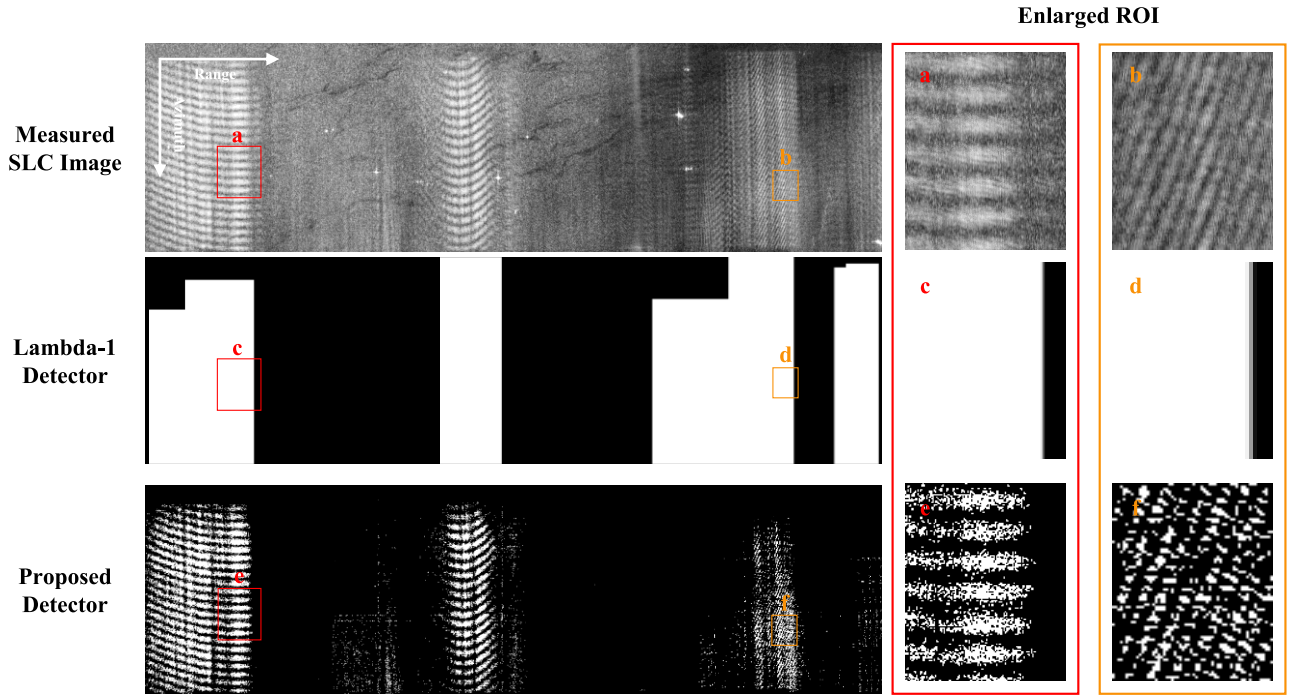


Fig. 24. Detection result of RFI artifacts based on Lambda-1 and the proposed detector in the Doha surrounding area acquired on September 16, 2019.

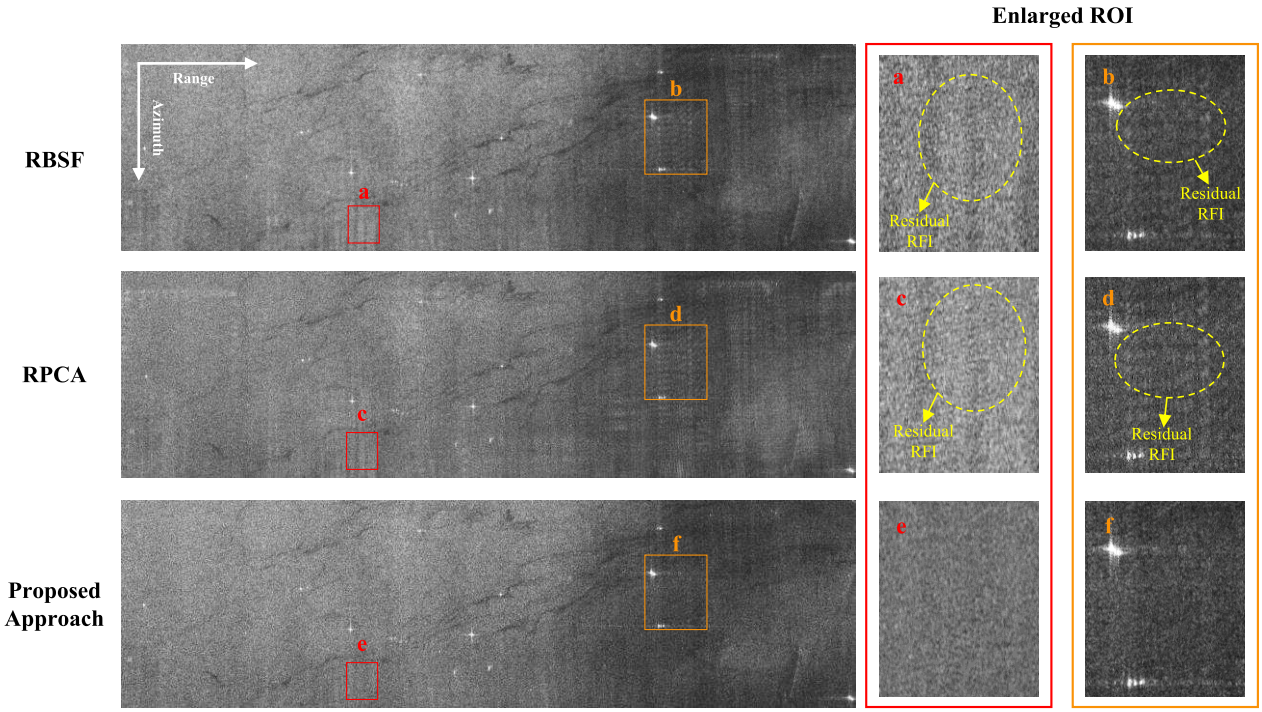


Fig. 25. RFI artifact mitigation result based on the RBSF, RPCA, and proposed approach.

further confirm that ASSC-RBSF is not merely an incremental improvement but a synergistic framework that overcomes the inherent limitations of individual SSC or RBSF, thus providing more reliable RFI mitigation performance for practical engineering applications.

It is worth noting that the approximate estimation of the ground-truth amplitude still exhibits slight discrepancies from actual observations. In addition, when the artifact energy is extremely weak, its features may be overwhelmed by the background, complicating RFI removal. Meanwhile, when RFI energy is excessively strong, some background details may be lost during mitigation. In future work, we will explore the integration of spatial-spectral features [55], [56] and learning-based [57] RFI detectors to improve artifact discrimination in both high- and low-SIR scenarios. Furthermore, we will further investigate high-performance mitigation strategies by

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incorporating more robust amplitude estimation techniques, such as robust PCA [58], [59] and multilook averaging [60], to enhance the robustness of the proposed RFI mitigation method under diverse RFI conditions.

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